

Qualifying Exam

November 7, 2025

Sam Buxbaum

Motivation

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Goal: Fill in the gaps that prevent the use of MPC in practice

Scaling Up MPC

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2. Directly incorporating more parties

- Significant practical challenges
- Distribute trust more widely

The Vision

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Part 1: For a fixed system model, improve the efficiency of a complex building block.

Part 2: For a moderately complex workload, scale the number of parties.

Roadmap

1. Oblivious Shuffle Preprocessing
2. Scaling with Many Parties

Oblivious Shuffle Preprocessing

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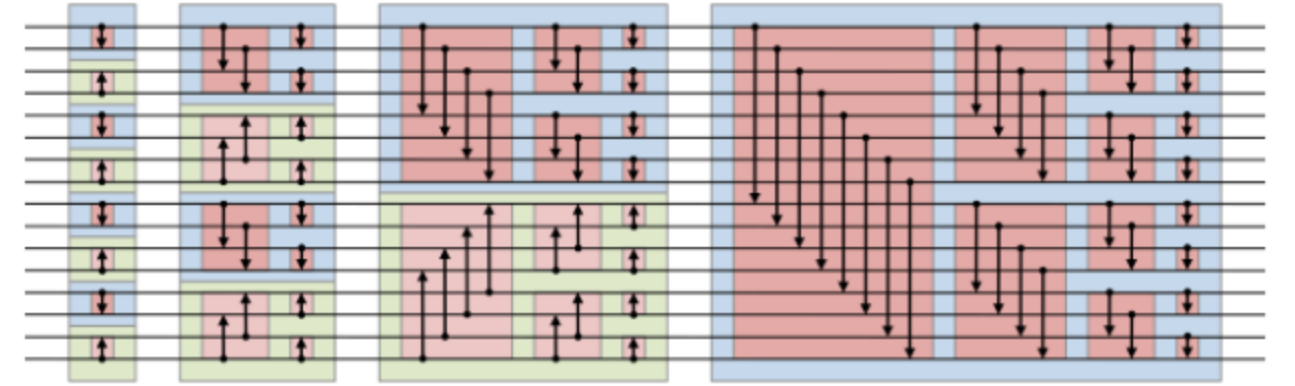
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- The output is a secret sharing $[\pi(\mathbf{x})]$

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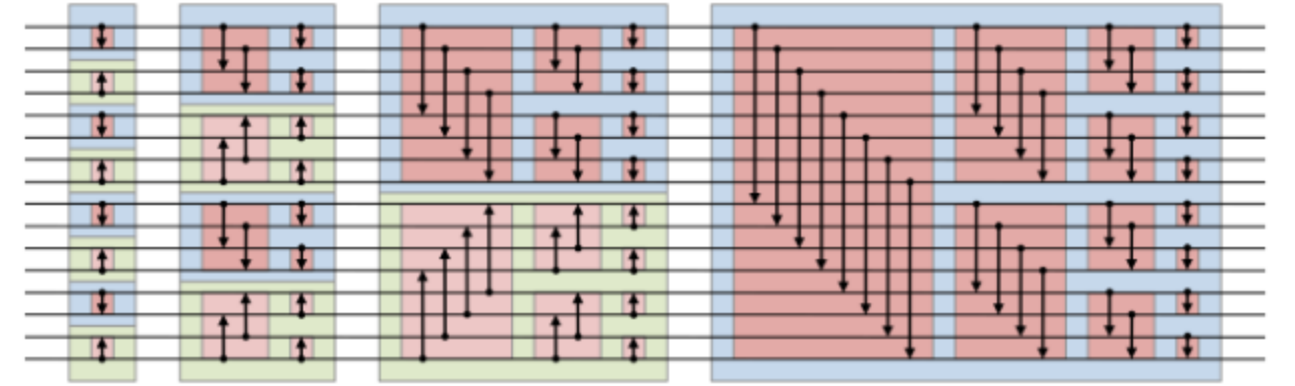
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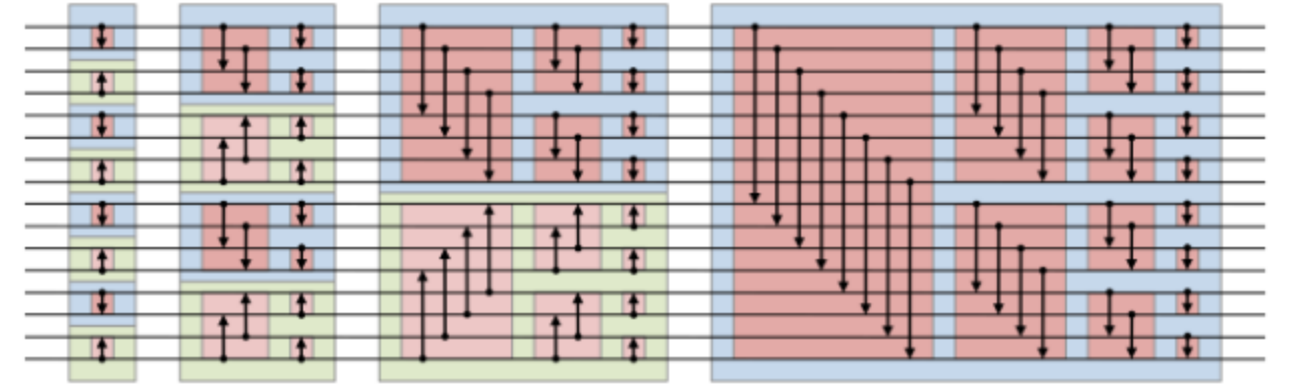


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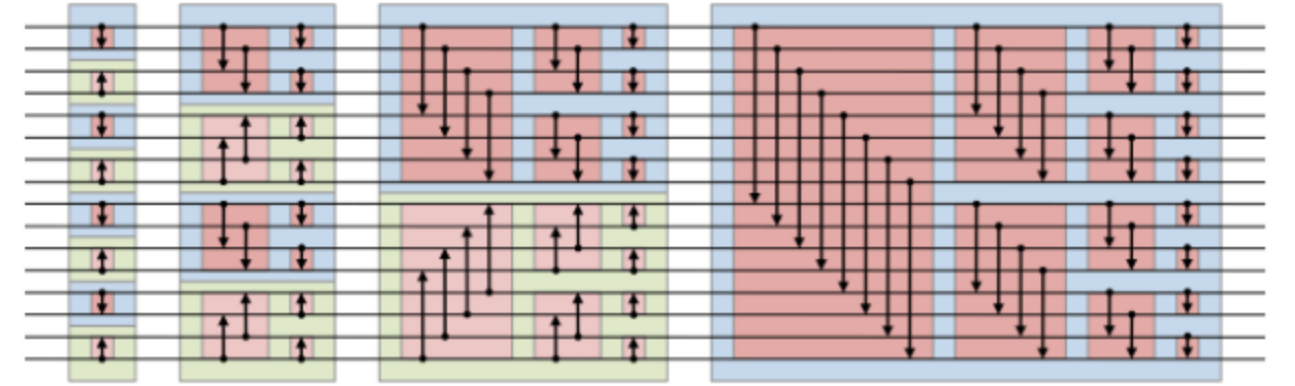
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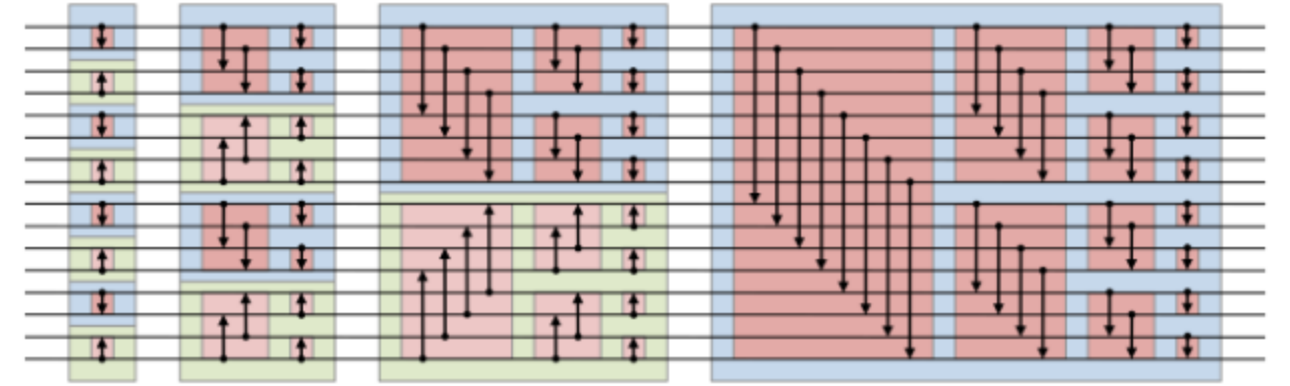
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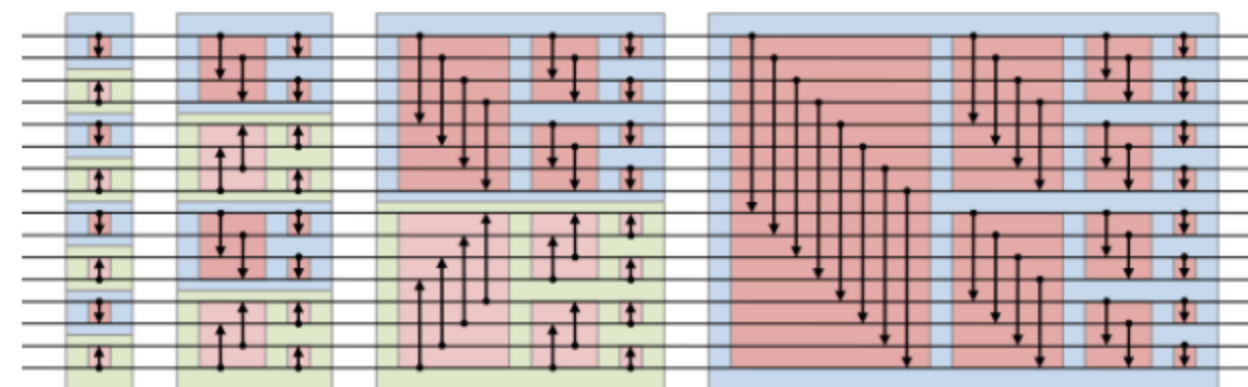
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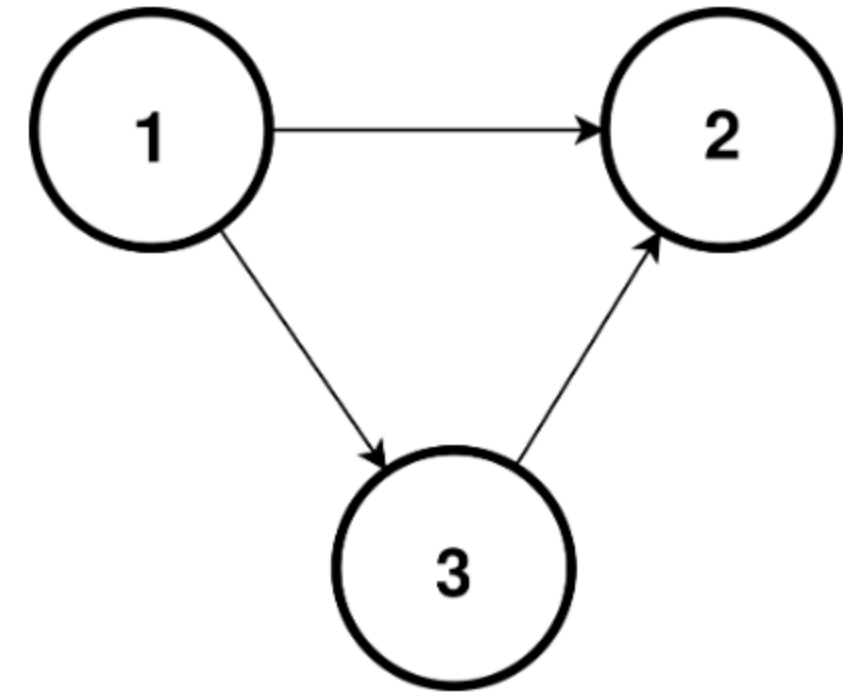
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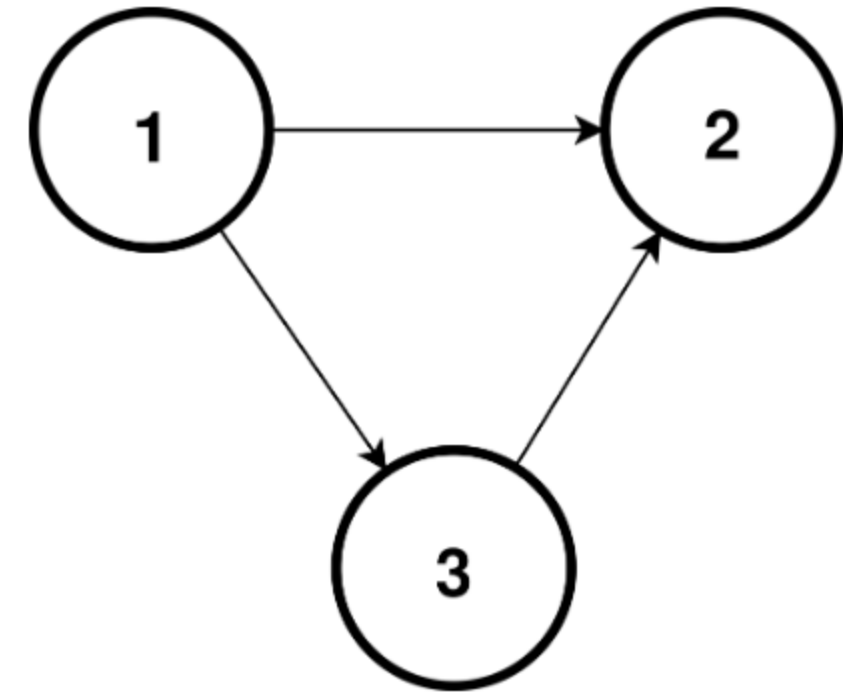
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- The revealed permutation is uncorrelated with the input
- Achieves $O(n \log n)$ secure comparisons

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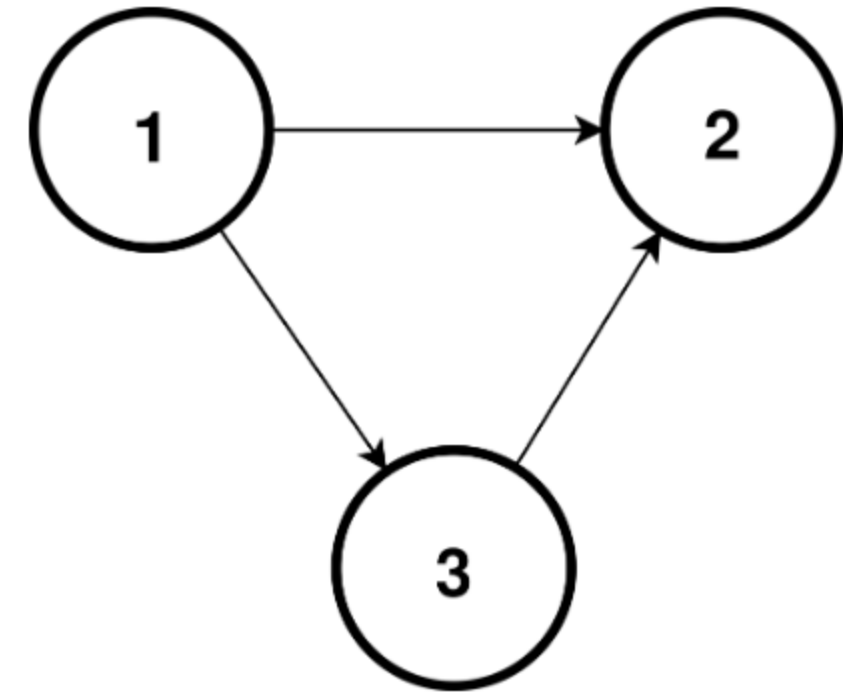
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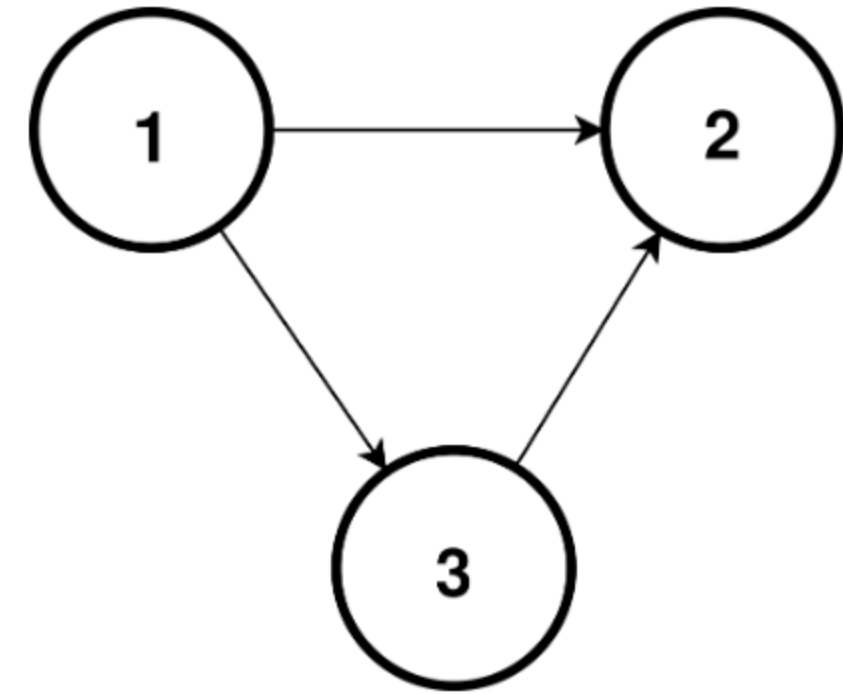


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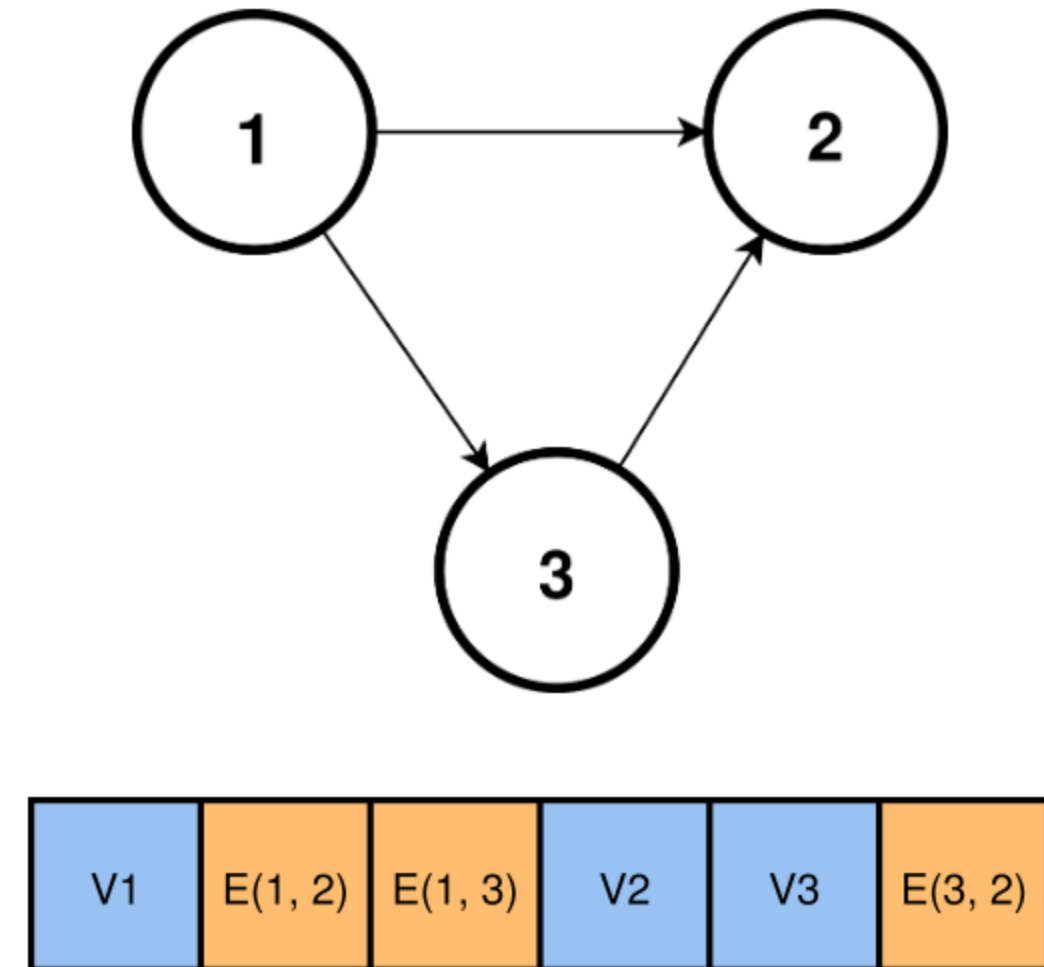
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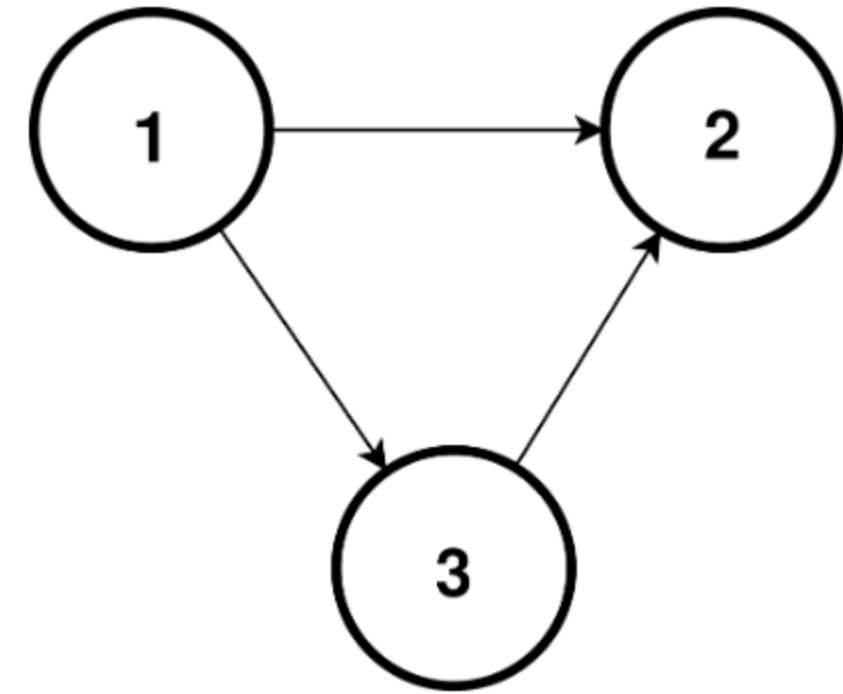
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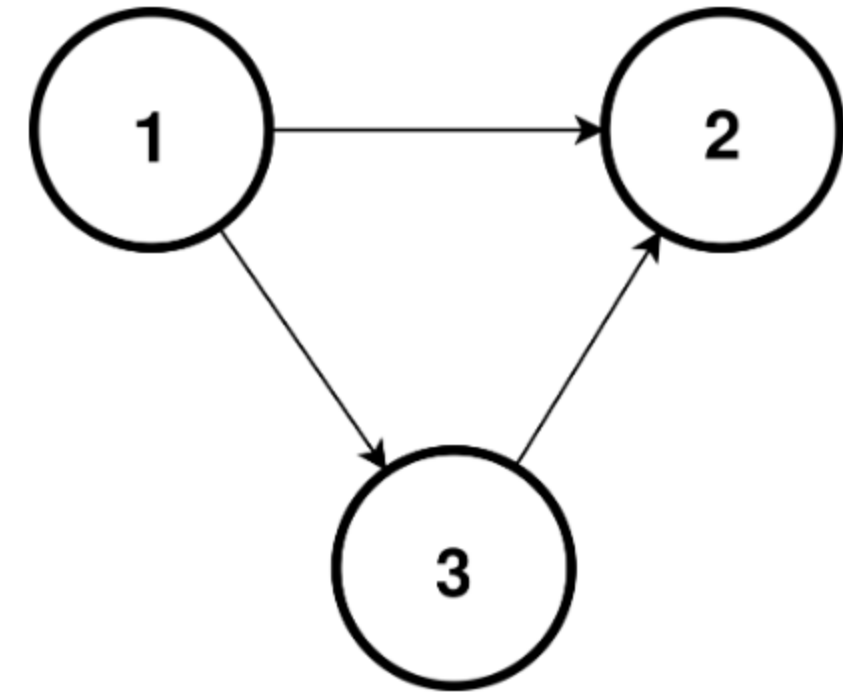
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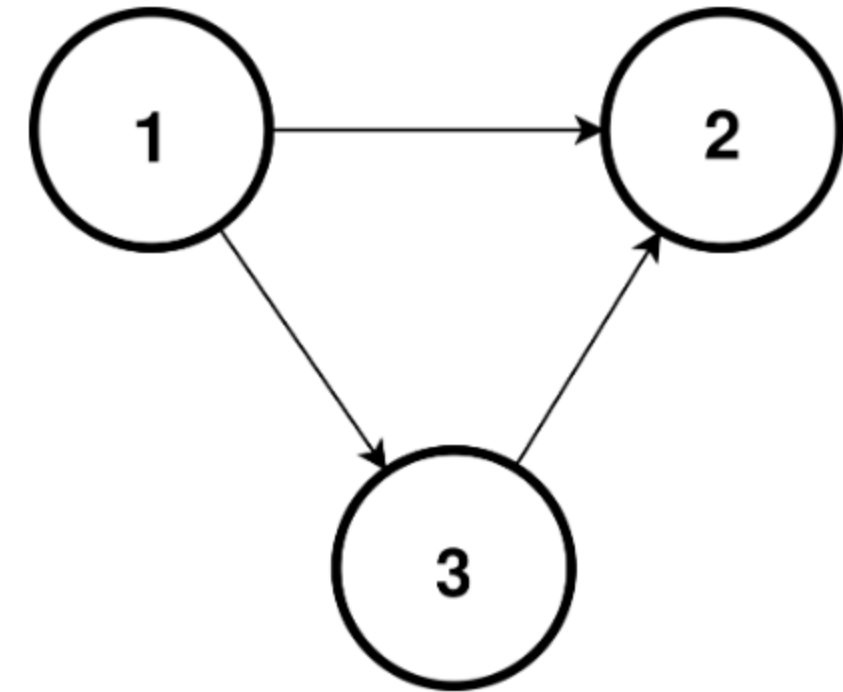
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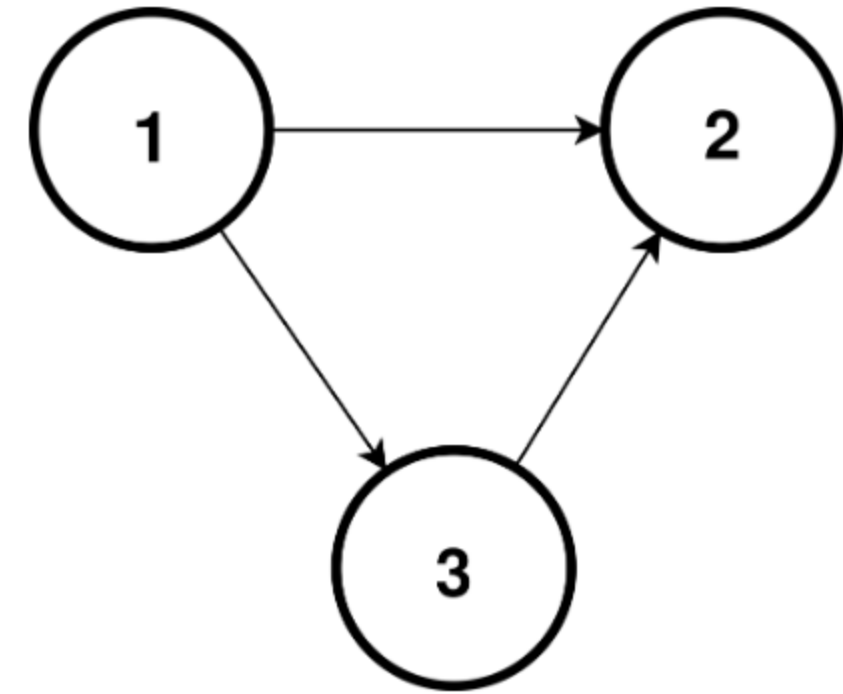
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- Shuffling allows us to convert between the orderings



Broader Applications

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How do we do it?

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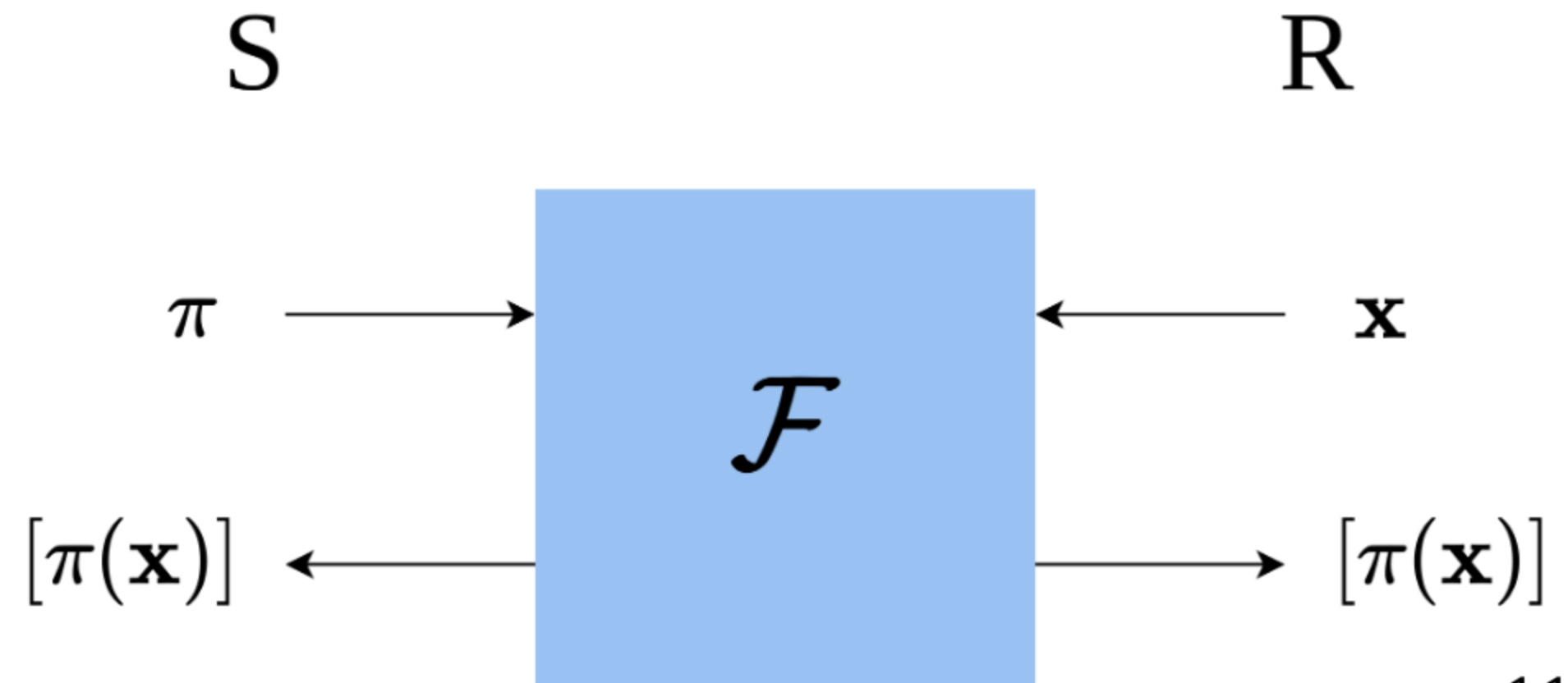
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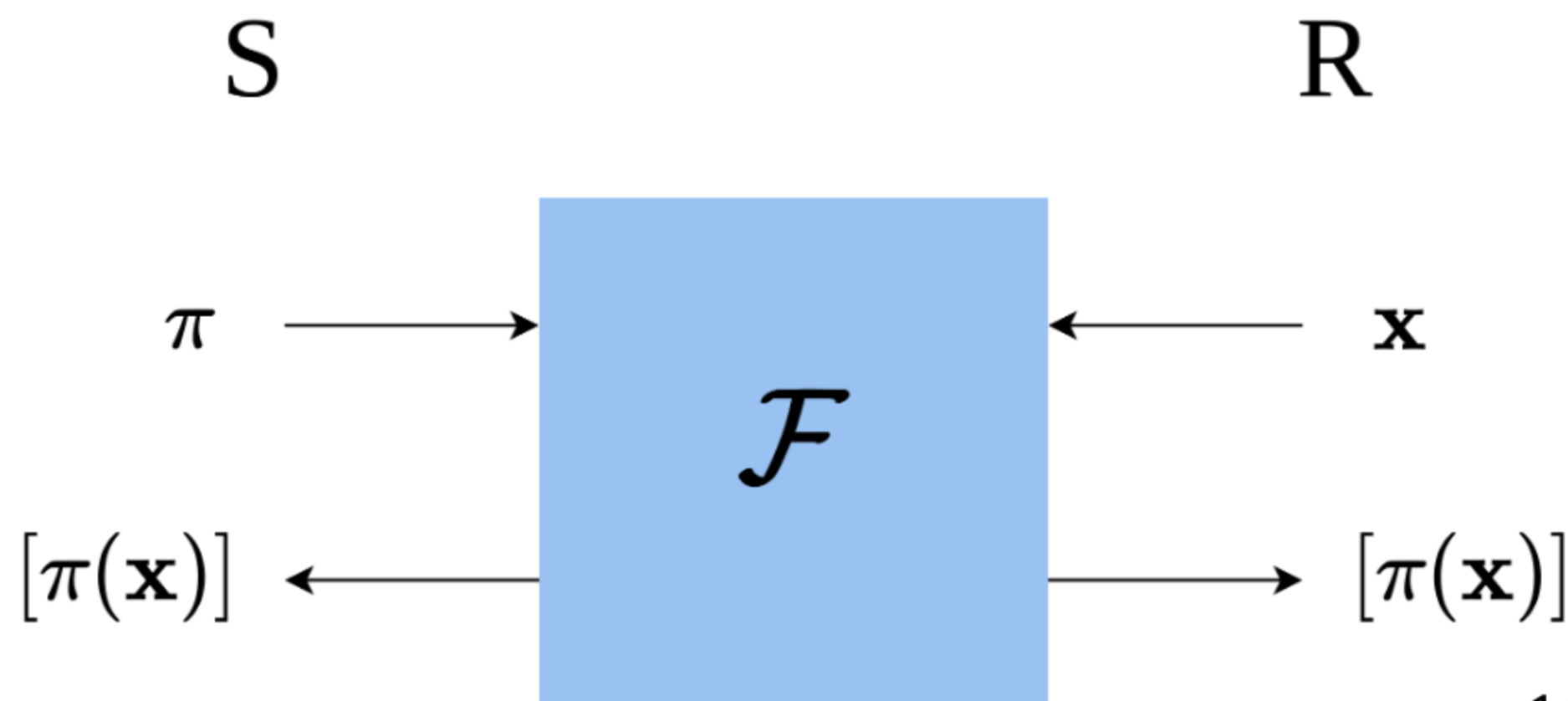
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- One party acts as a sender, the other as a receiver
- Known as "Permute+Share" (Chase et al.)



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$$(\pi, C), (A, B)$$

$$\pi(A) = B + C$$

Online Derandomization Protocol

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S: (π, C)

R: $(\mathbf{x}, (A, B))$

$\mathbf{x} - A$



$\pi(\mathbf{x} - A) + C$

B

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S: (π, C)

R: $(\mathbf{x}, (A, B))$

$\xleftarrow{\mathbf{x} - A}$

$\pi(\mathbf{x} - A) + C$

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$$\begin{aligned}\text{output} &= \pi(\mathbf{x} - A) + C + B \\ &= \pi(\mathbf{x} - A) + \pi(A) \\ &= \pi(\mathbf{x}) - \pi(A) + \pi(A) \\ &= \pi(\mathbf{x})\end{aligned}$$

Formally Defining the Problem

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Functionality $\mathcal{F}_{\text{GenPerm}}$

PUBLIC PARAMETERS: Number of elements n and bitwidth ℓ .

INPUT: None

OUTPUT: The functionality samples a uniformly random permutation $\pi \leftarrow S_n$ and uniformly random vectors $A, B, C \in \{0, 1\}^{n \times \ell}$, such that $\pi(A) = B + C$. $\mathcal{F}_{\text{GenPerm}}$ outputs (π, C) to the sender and (A, B) to the receiver.

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- p , the total number of permutation correlations generated

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Peceny, Raghuraman, Rindal, and Shah (PKC 2025)

Efficient Permutation Correlations and Batched Random Access for Two-Party Computation

Stanislav Peceny^{1*}, Srinivasan Raghuraman², Peter Rindal³, and Harshal Shah³

¹ Georgia Institute of Technology

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Abstract. In this work we formalize the notion of a two-party permutation correlation $(A, B), (C, \pi)$ s.t. $\pi(A) = B + C$ for a random permutation π of n elements and vectors $A, B, C \in \mathbb{F}^n$. This correlation can be viewed as an abstraction and generalization of the Chase et al. (Asiacrypt 2020) share translation protocol. We give a systematization of knowledge for how such a permutation correlation can be derandomized

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Construction 1

Construction 2

$O(n\ell)$

$O(n \log \ell)$

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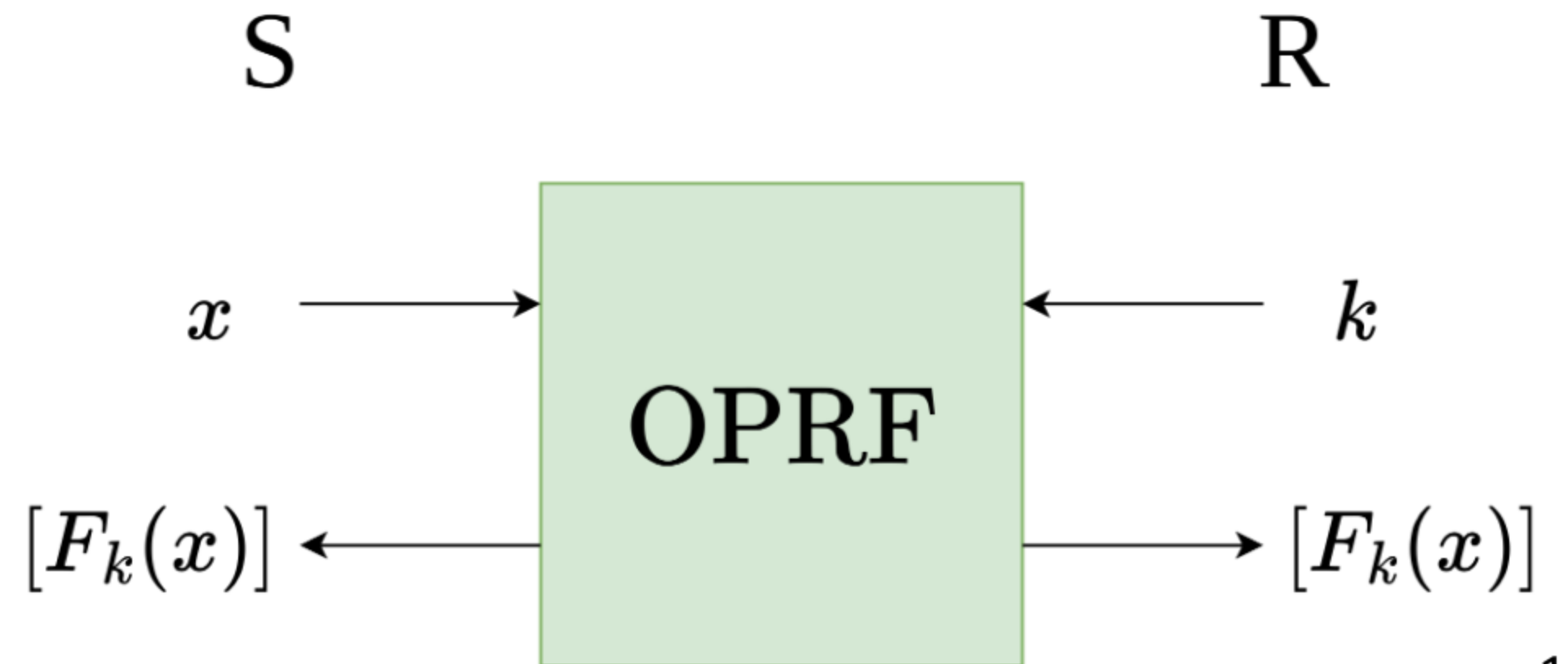
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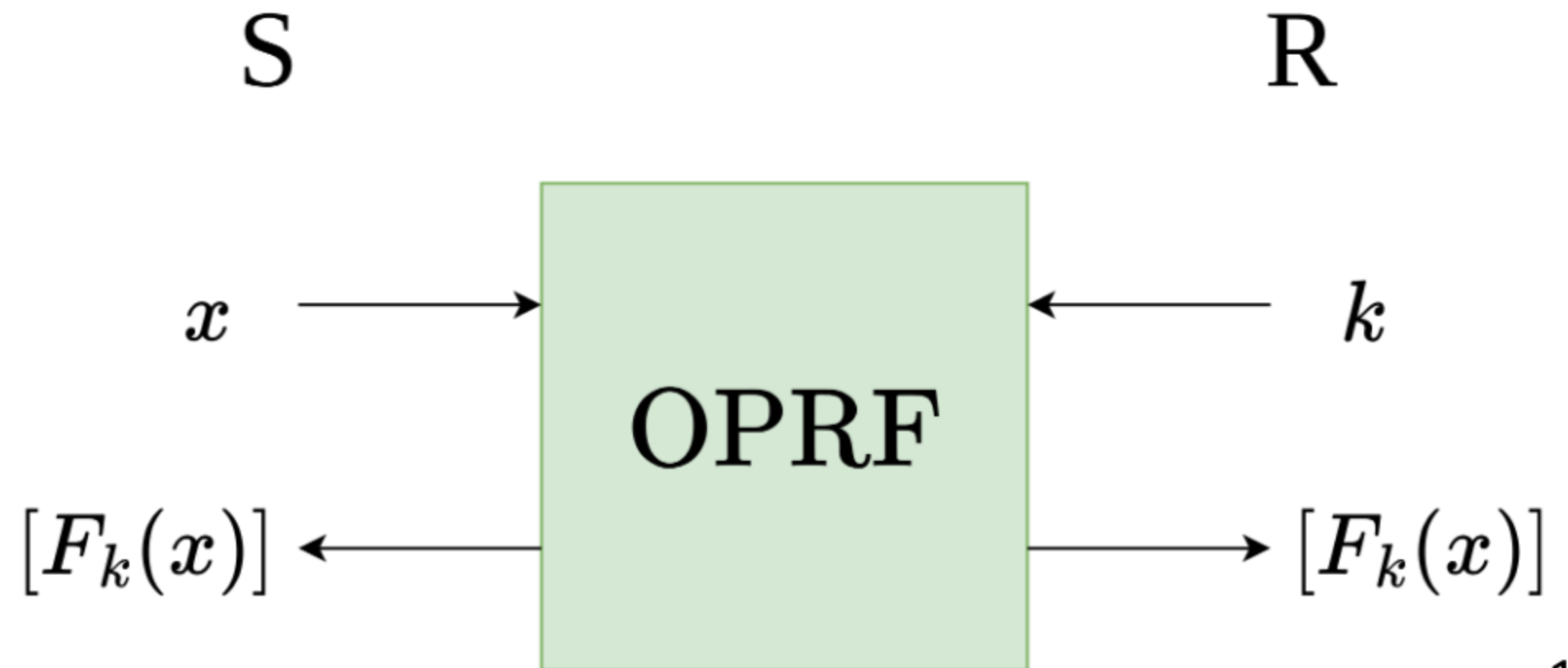
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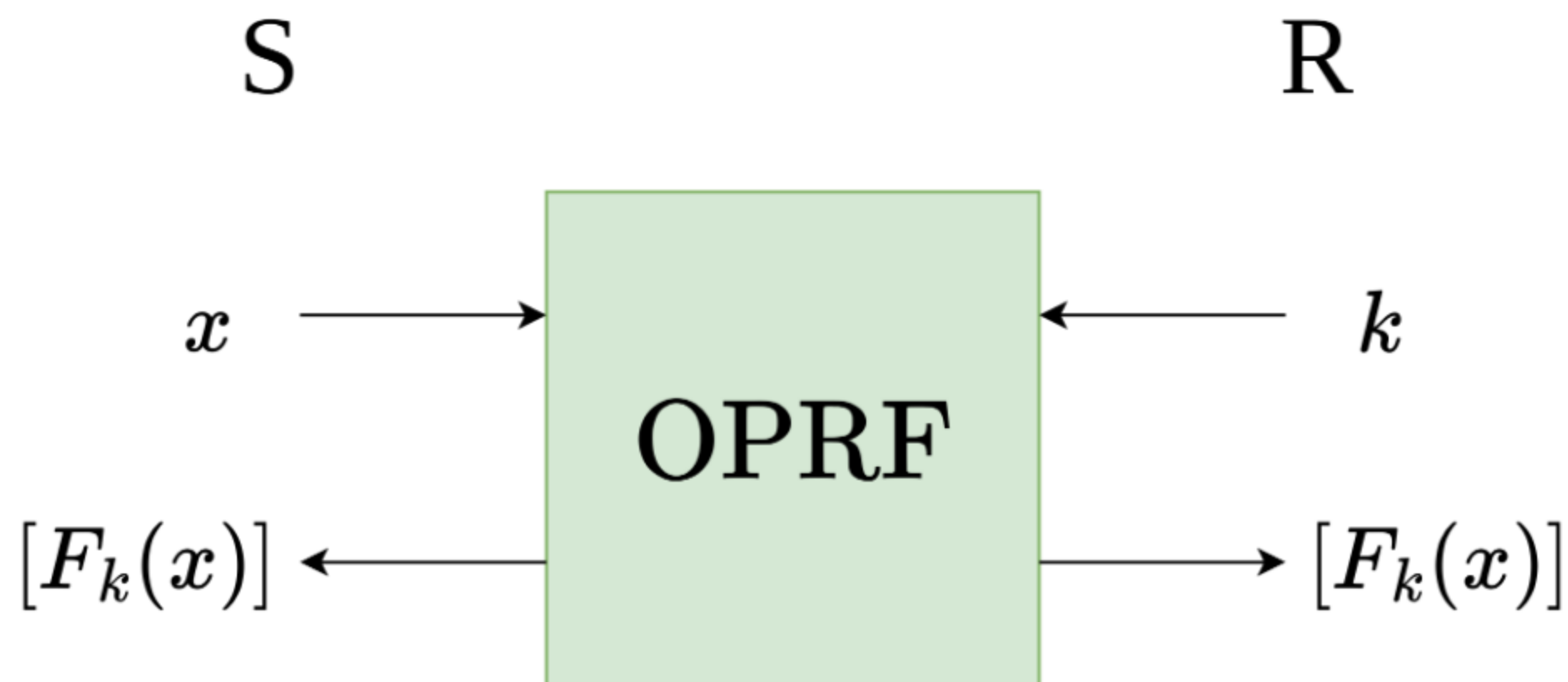
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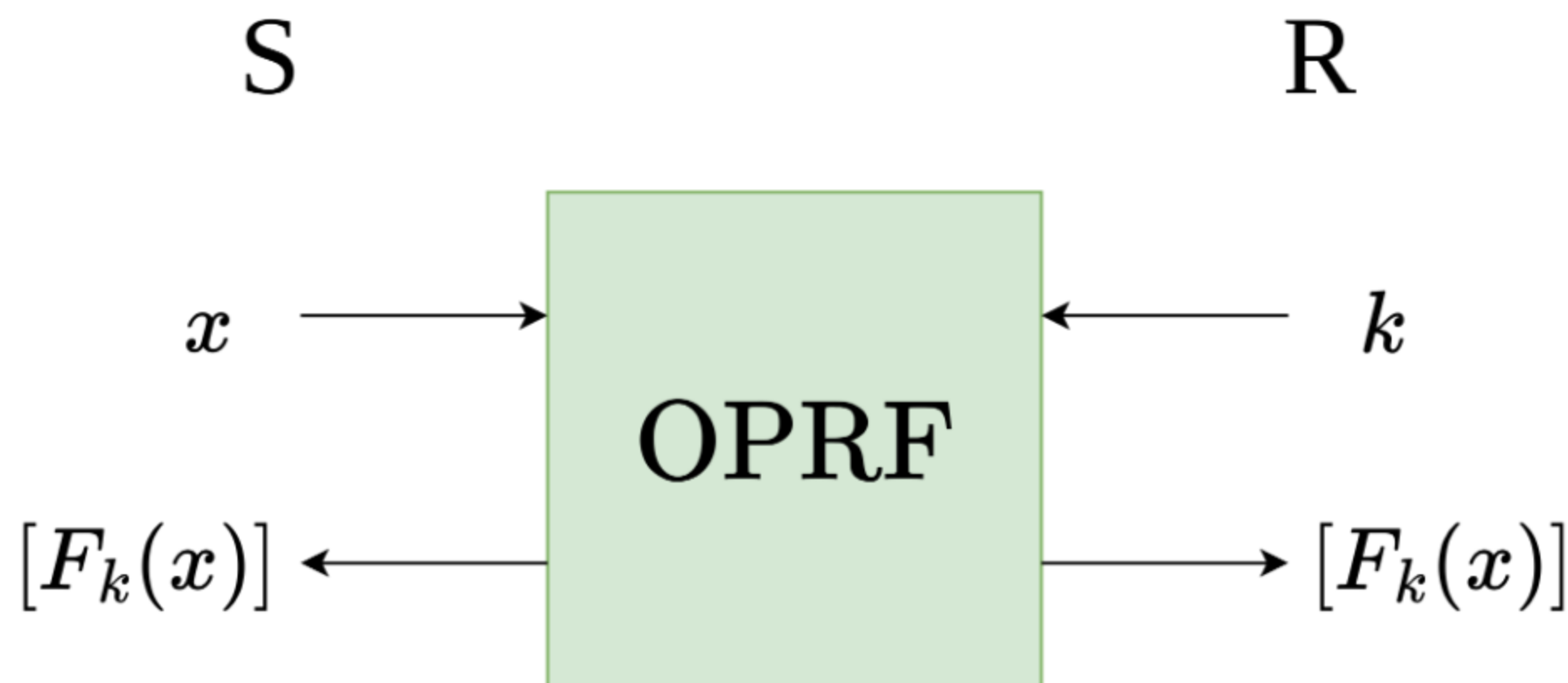
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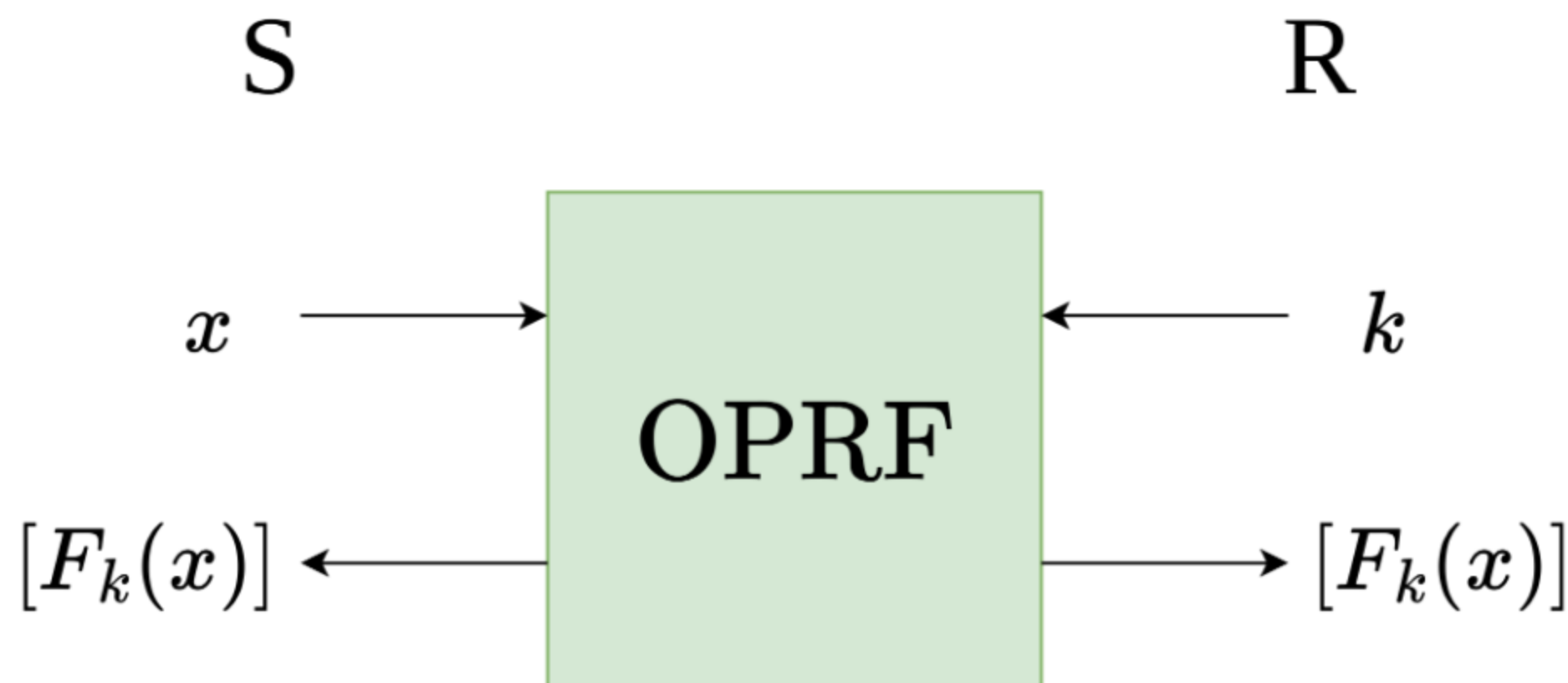
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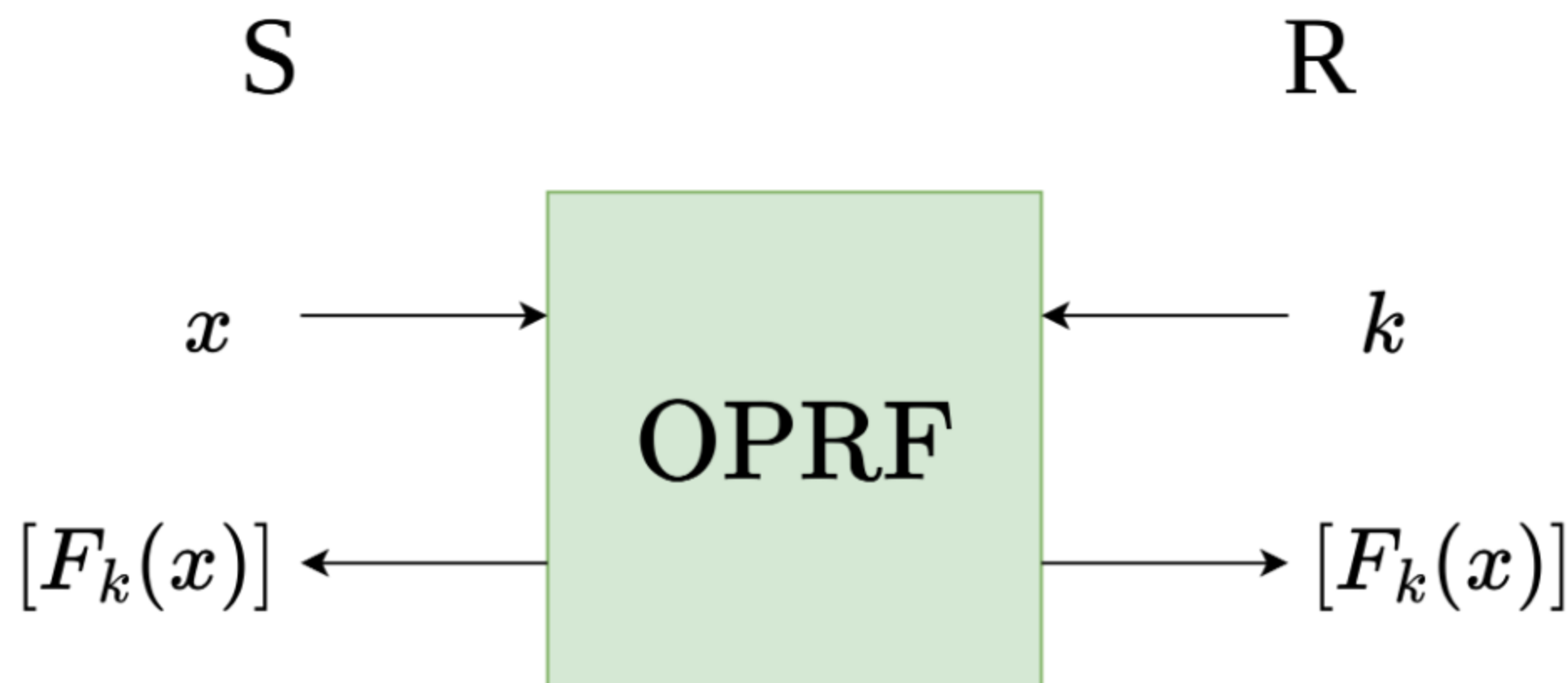
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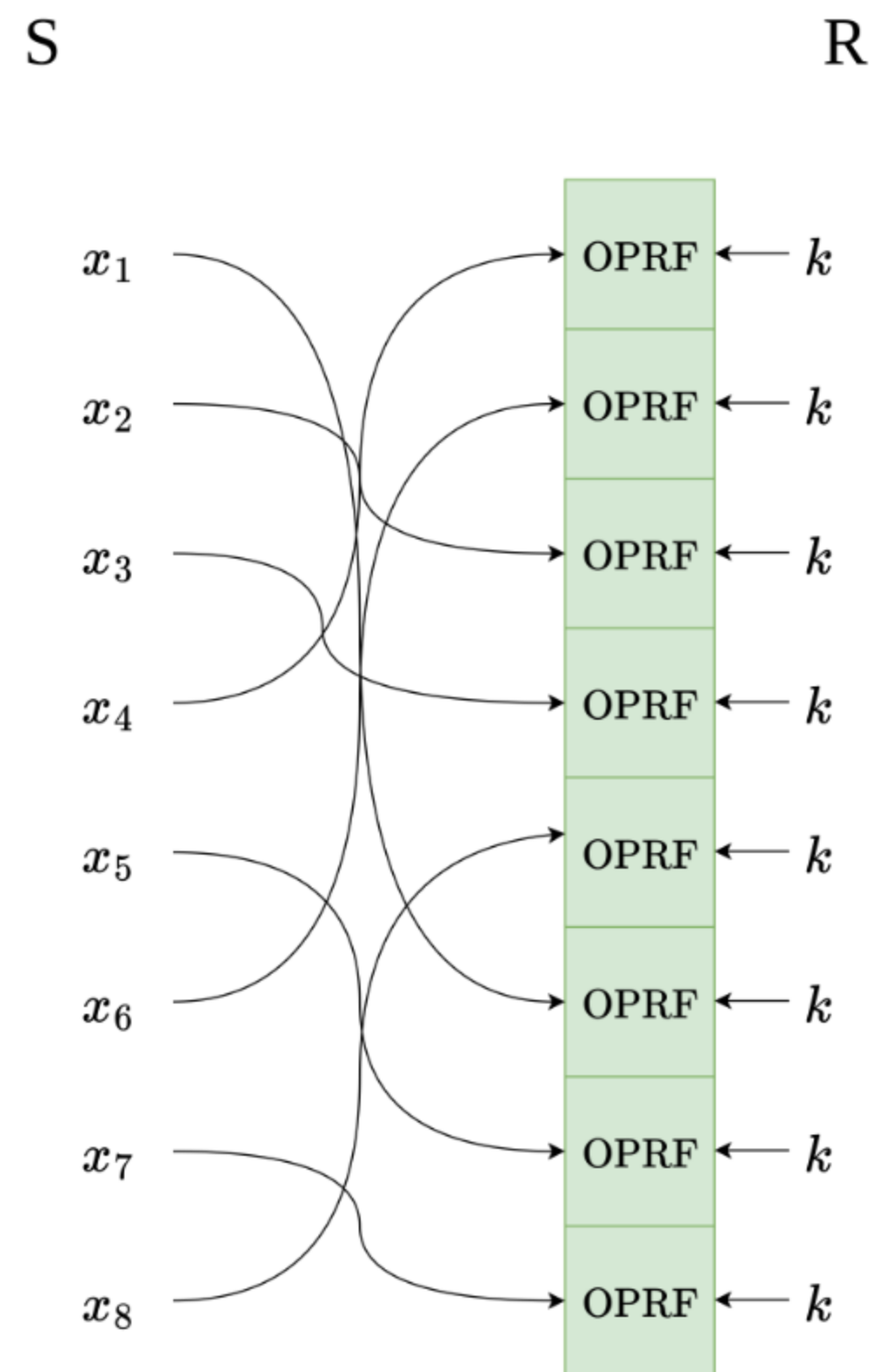
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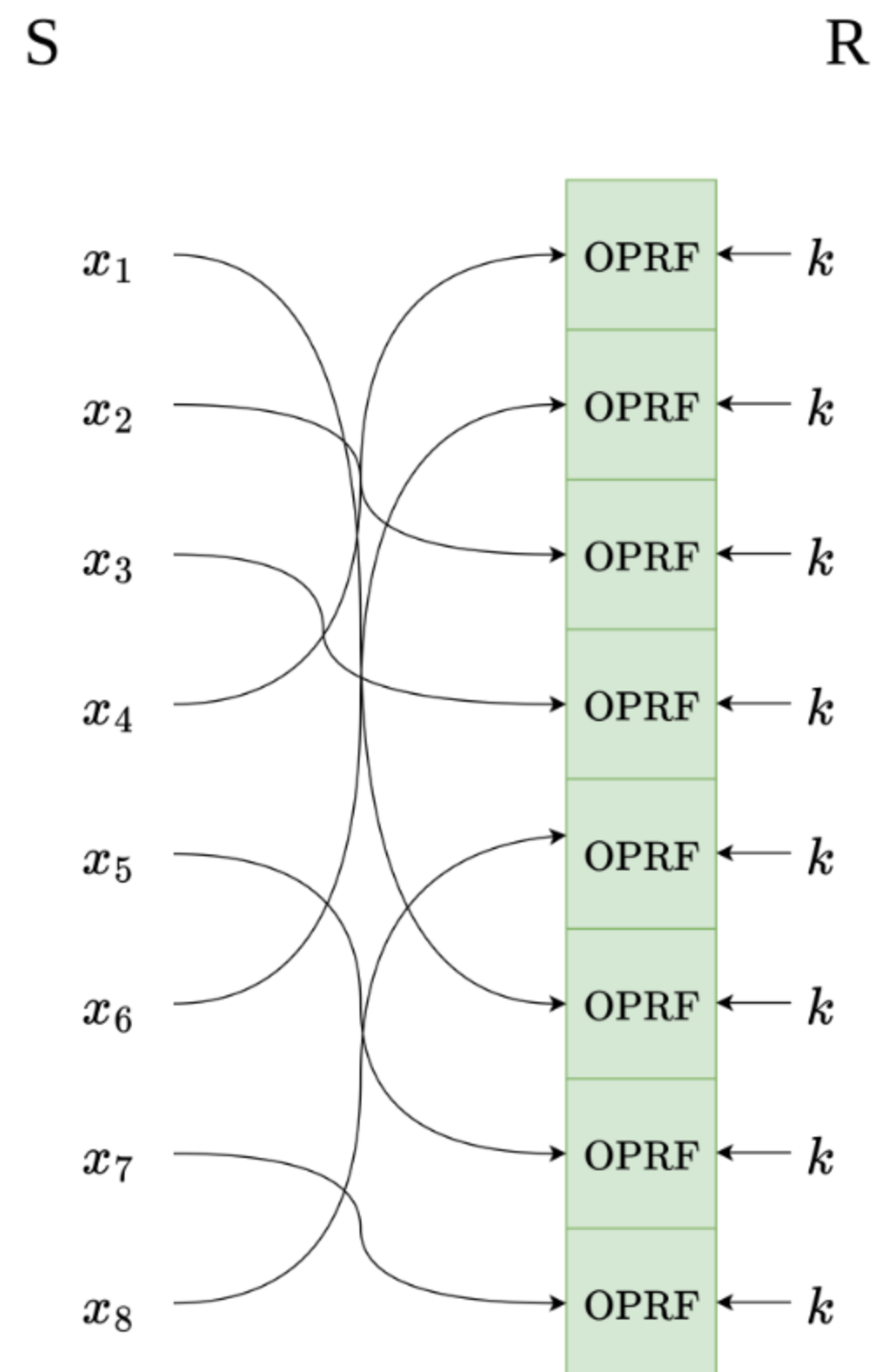


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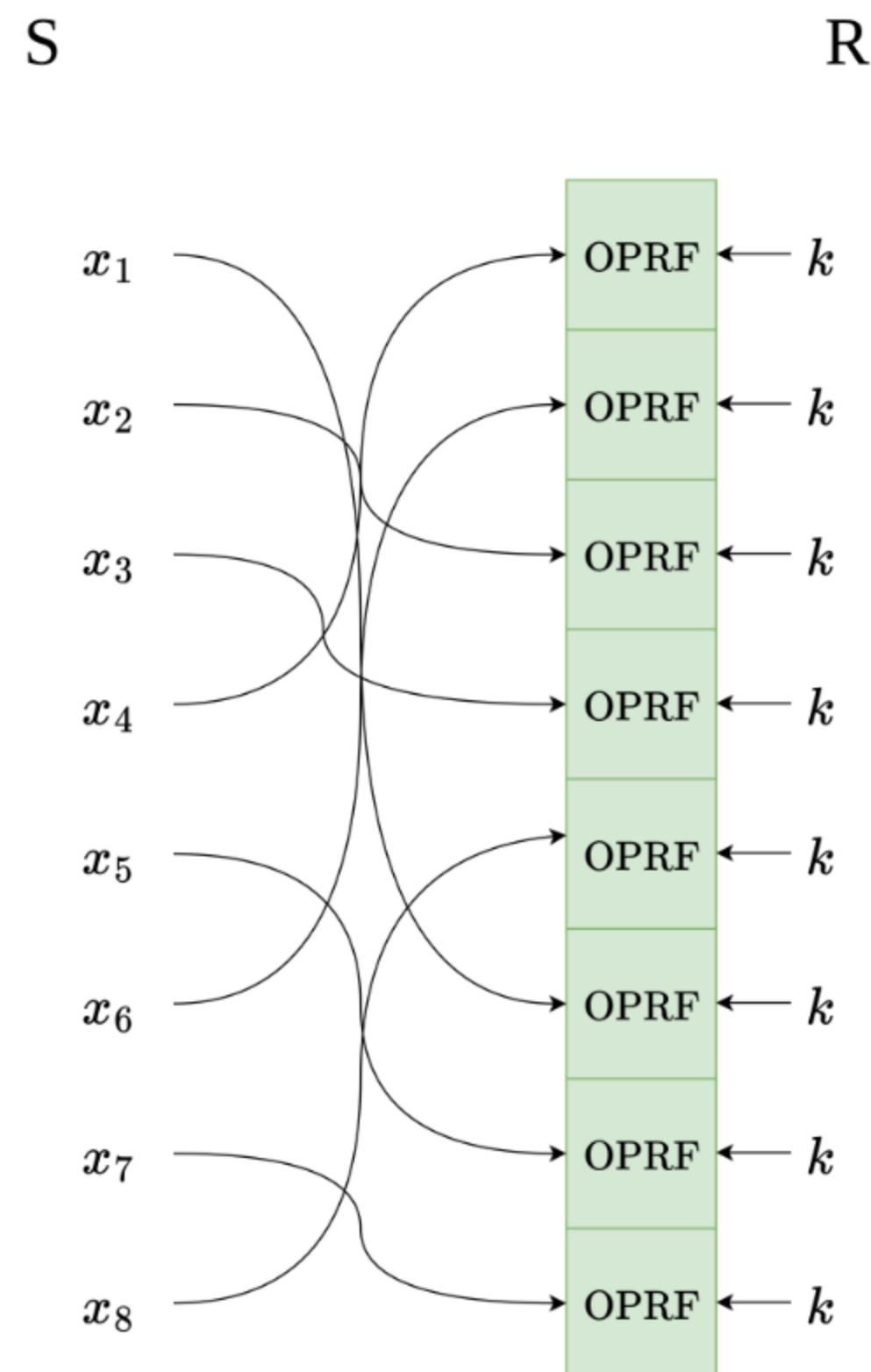
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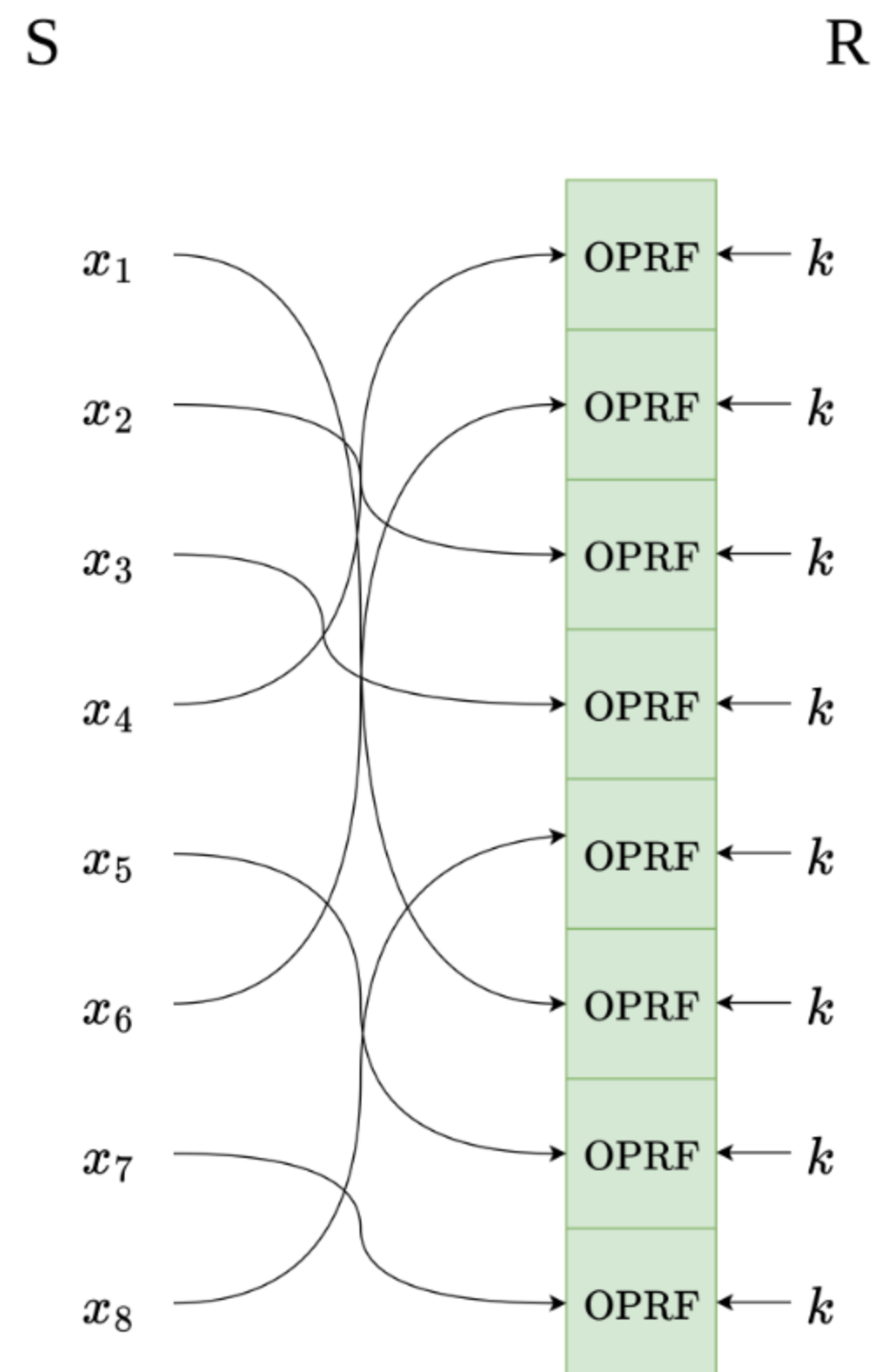
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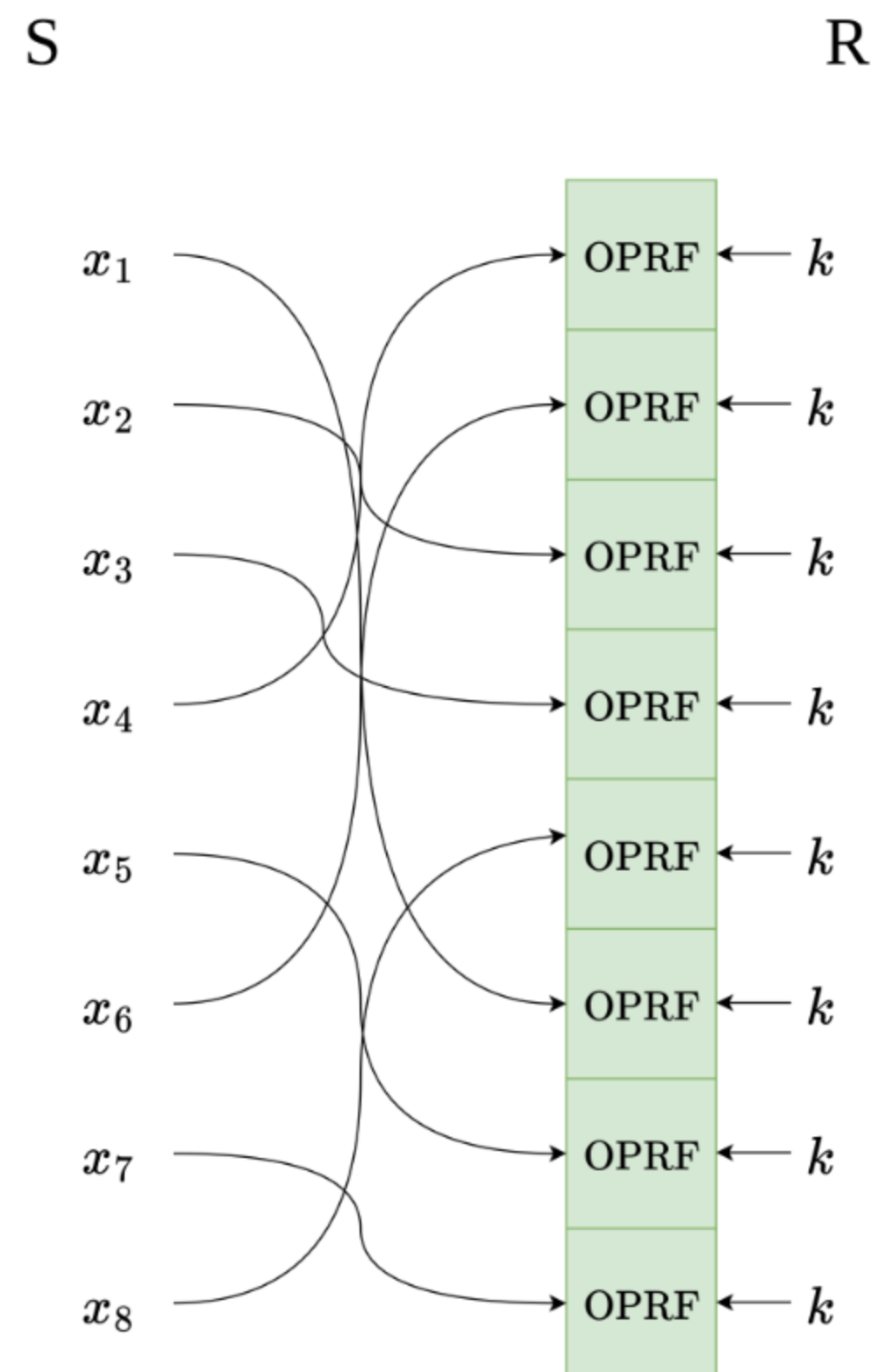
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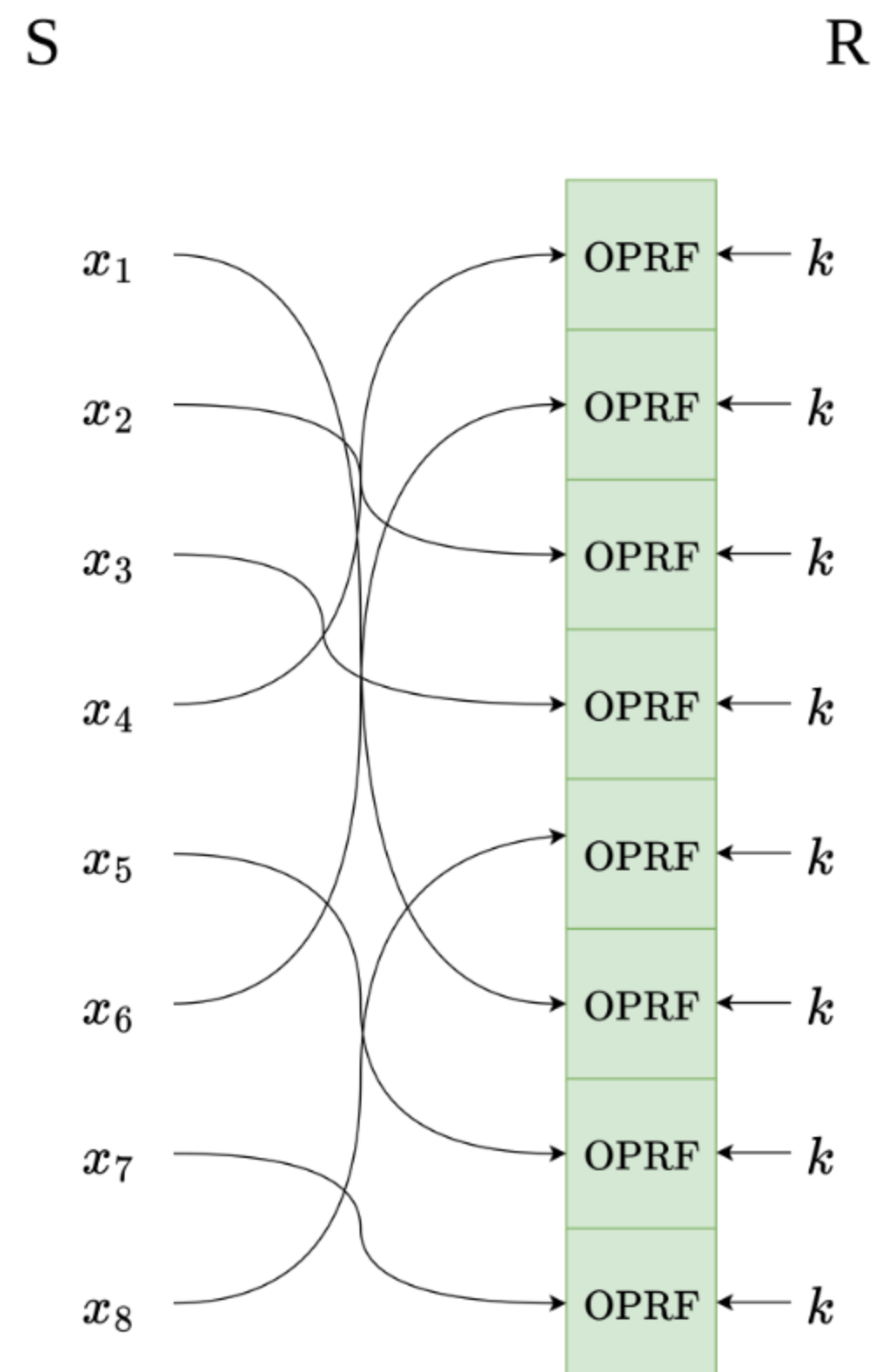
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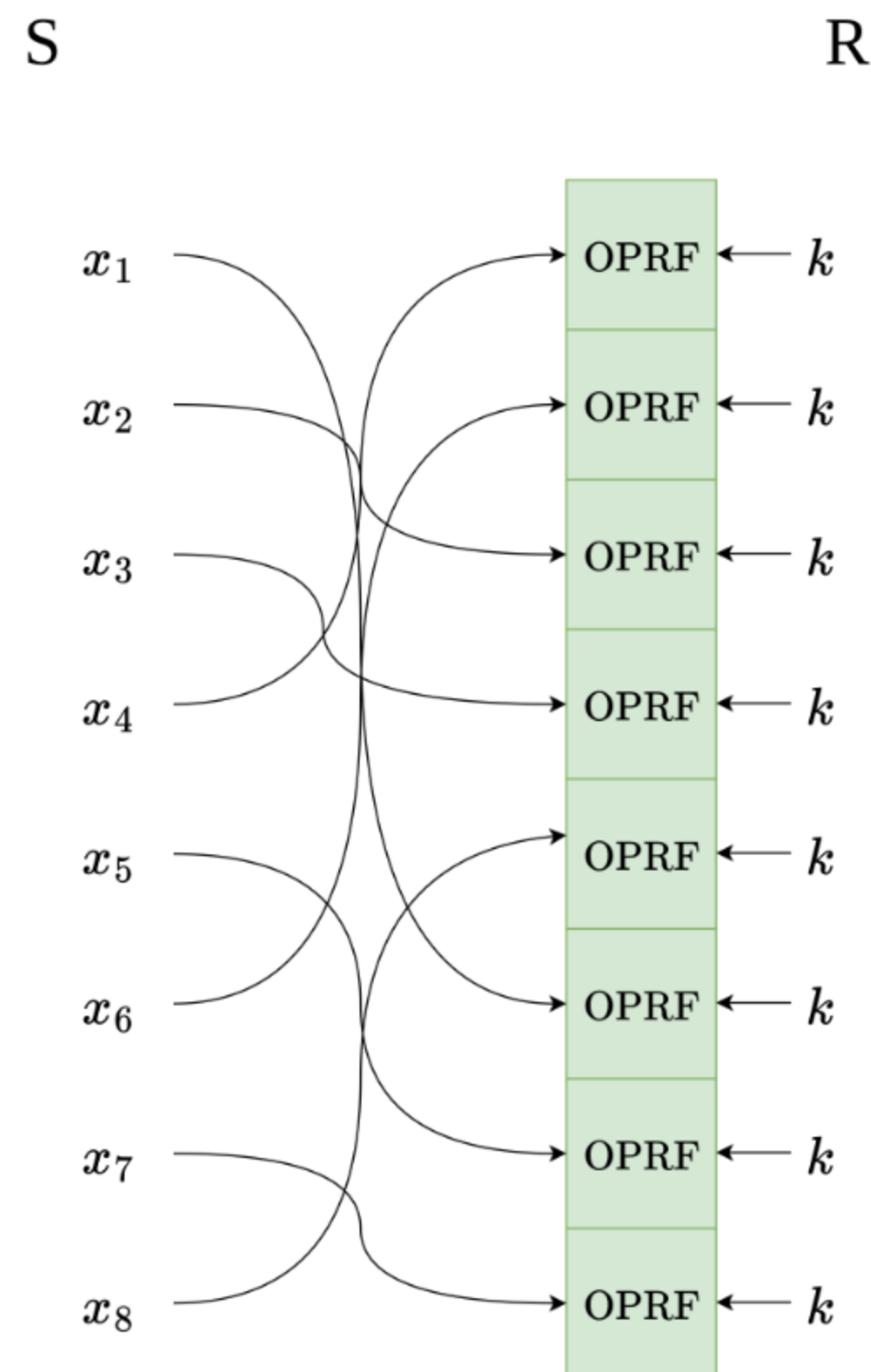
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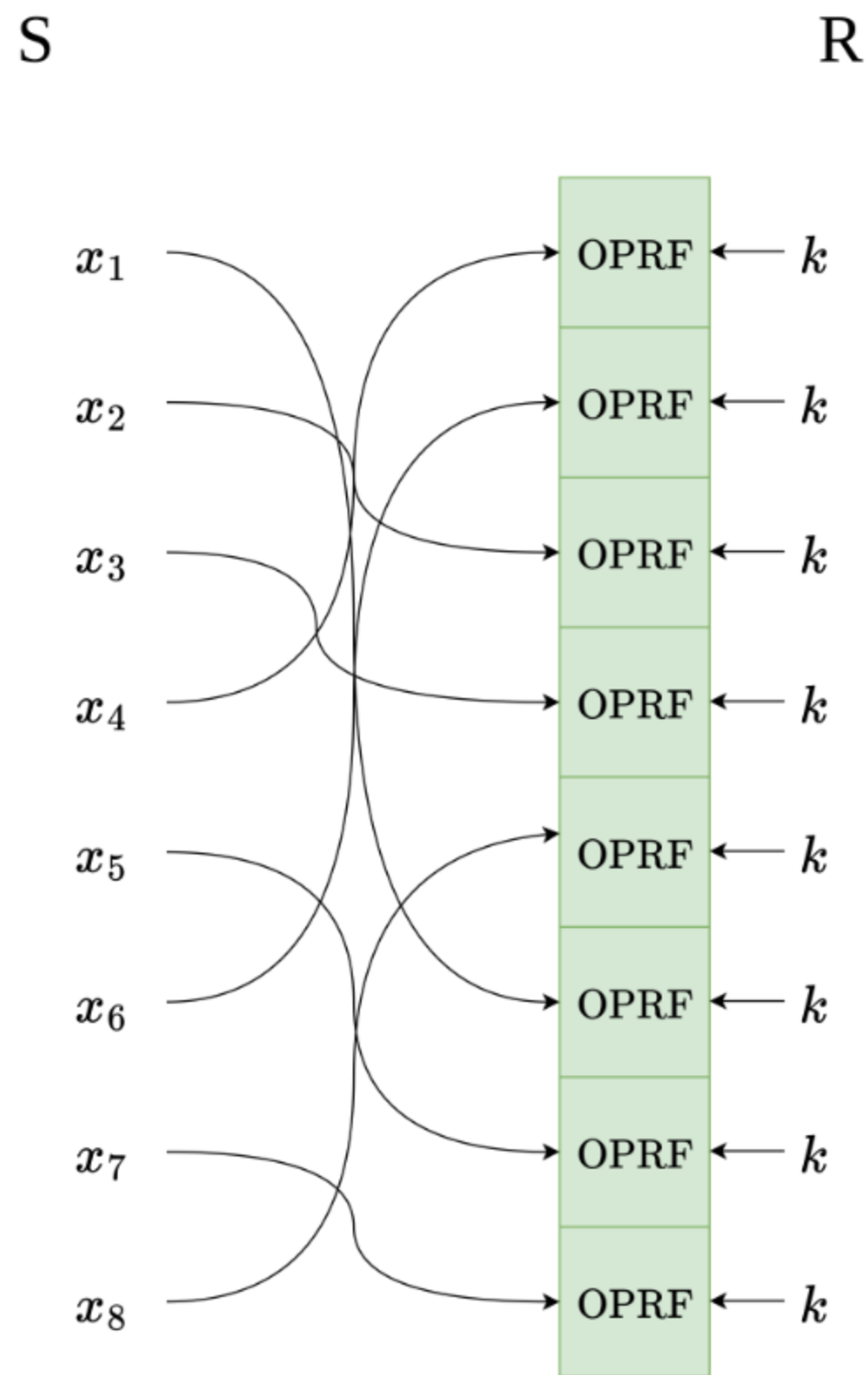


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- R sets $A_i = F_k(x_i)$ in plaintext
- Parties set B and C as the OPRF outputs

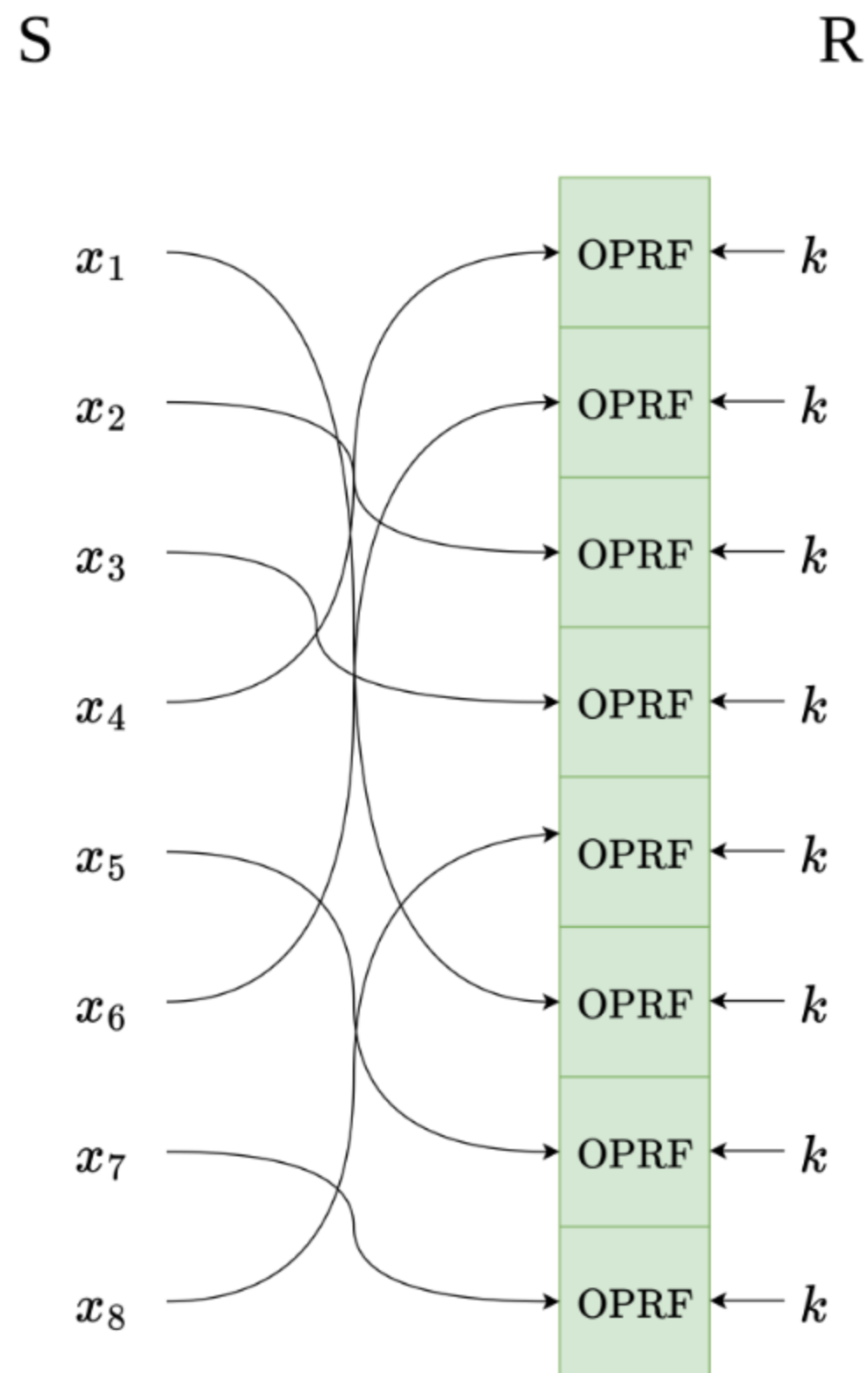


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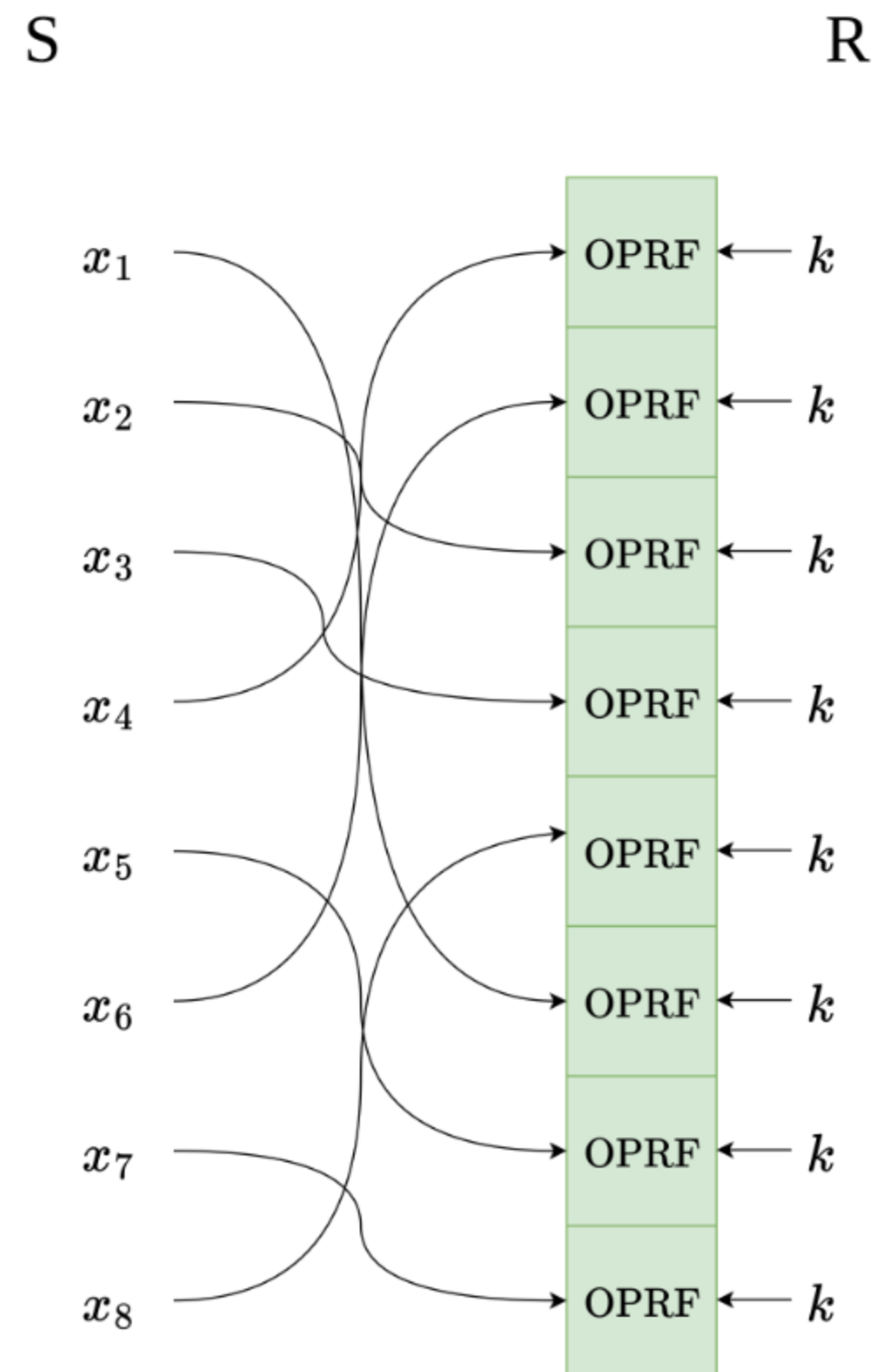
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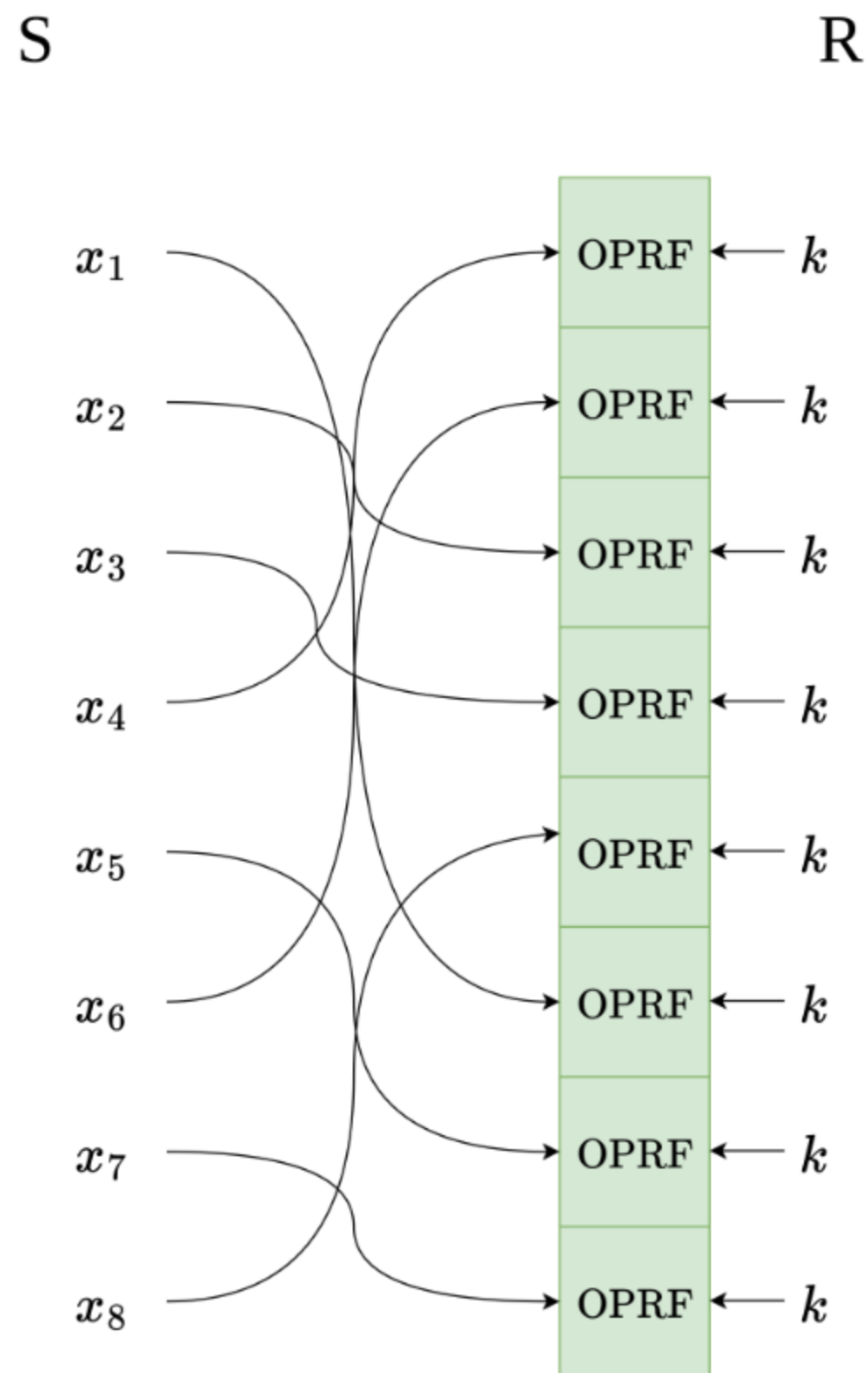
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- n calls to the OPRF
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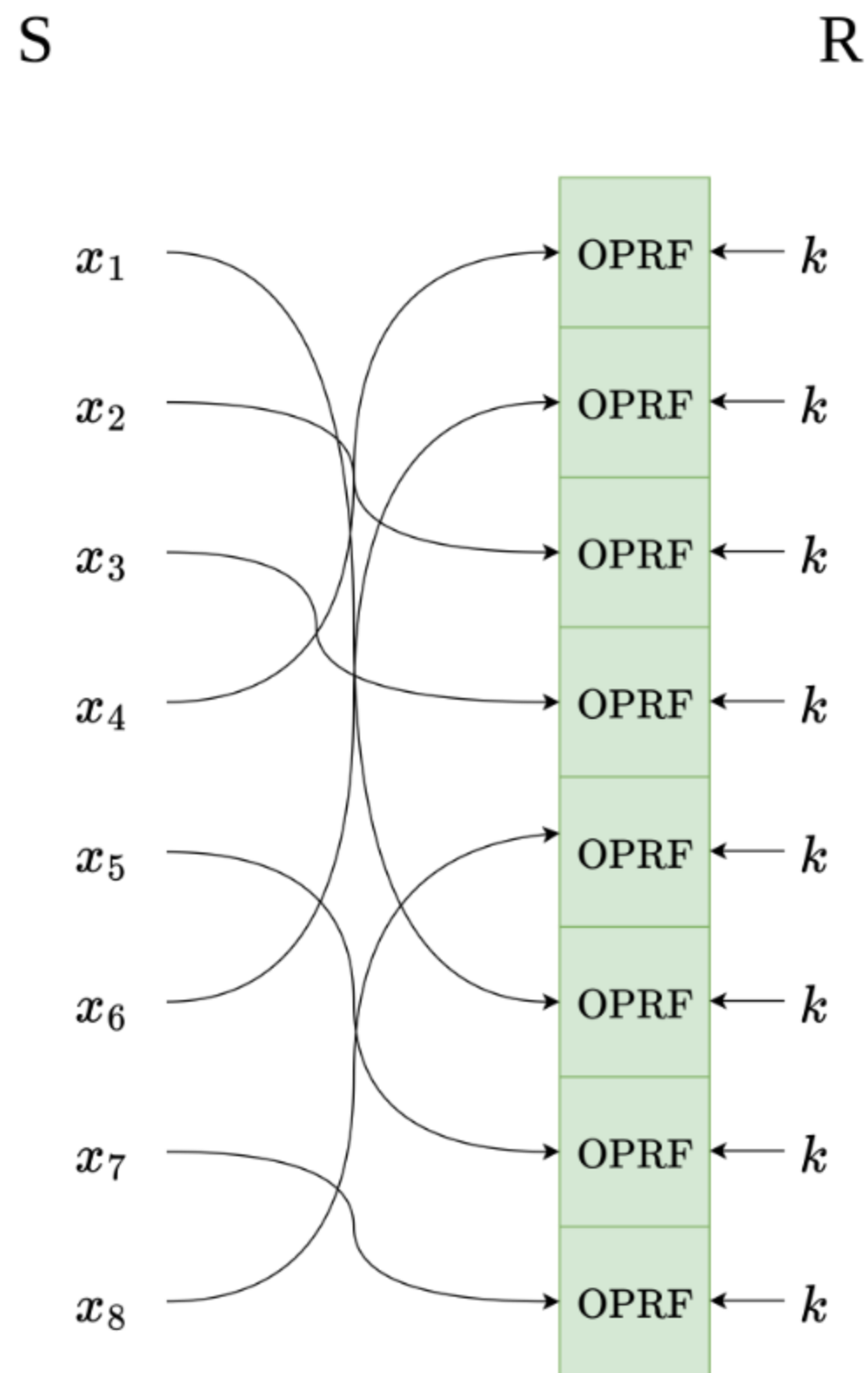
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- $\approx 4s$ per million elements



Sublinear Protocol

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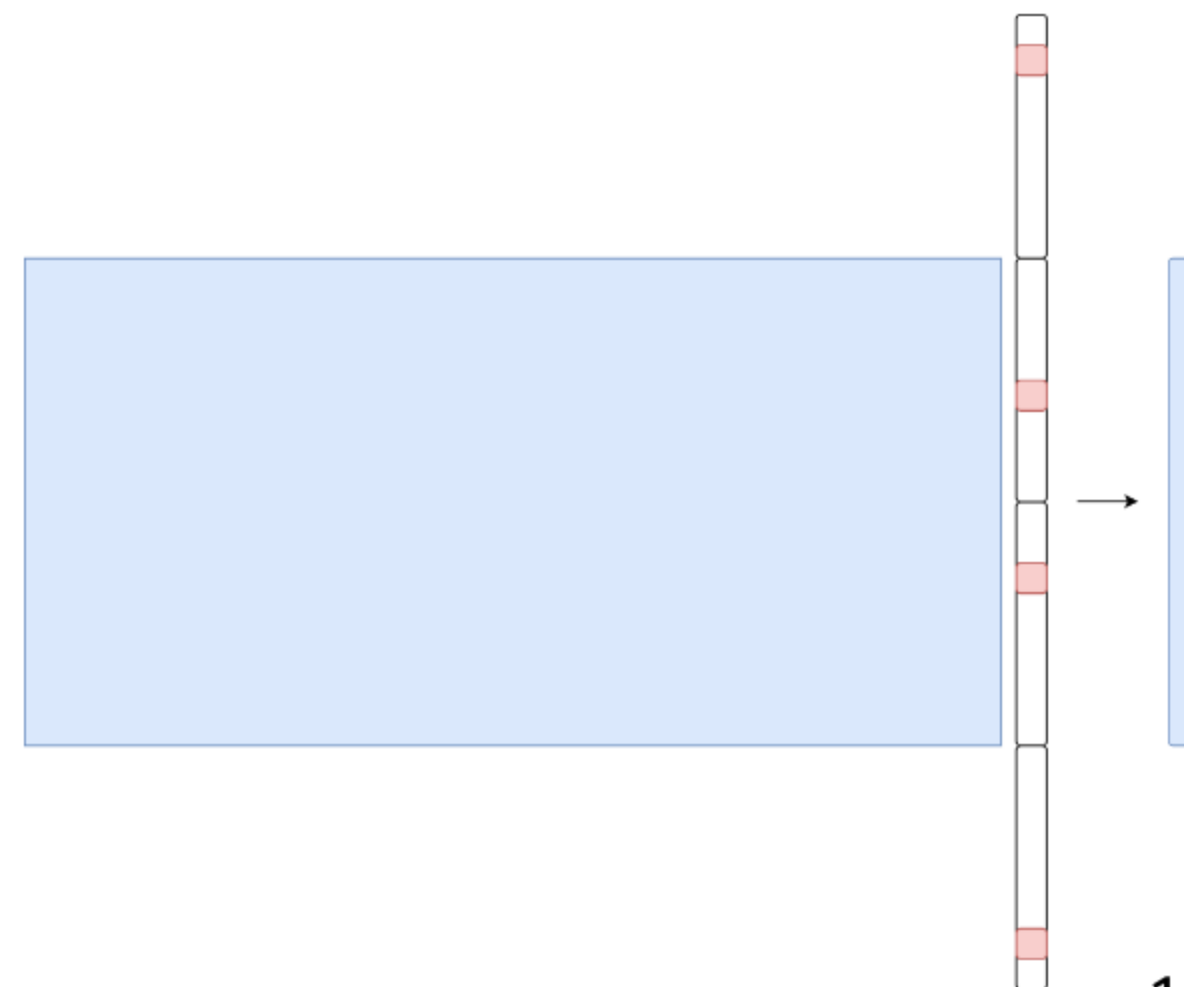
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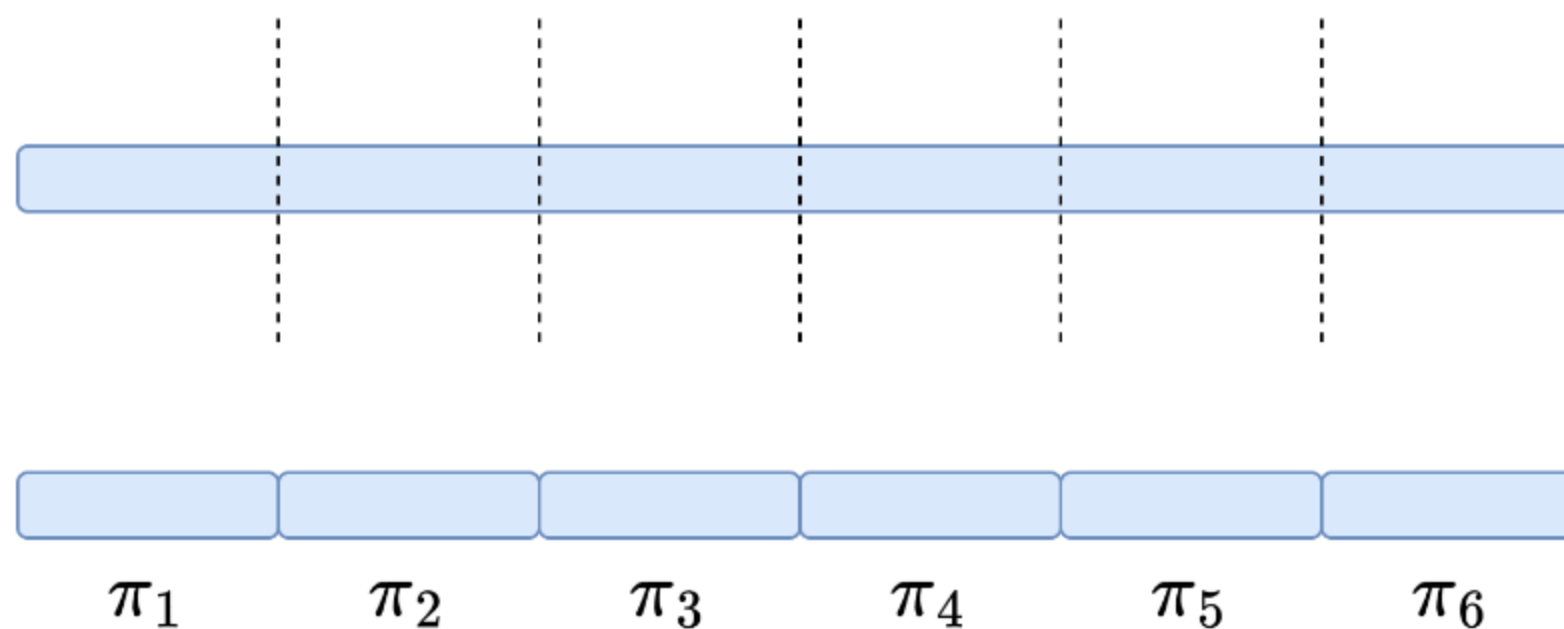
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Ongoing work: graph algorithms

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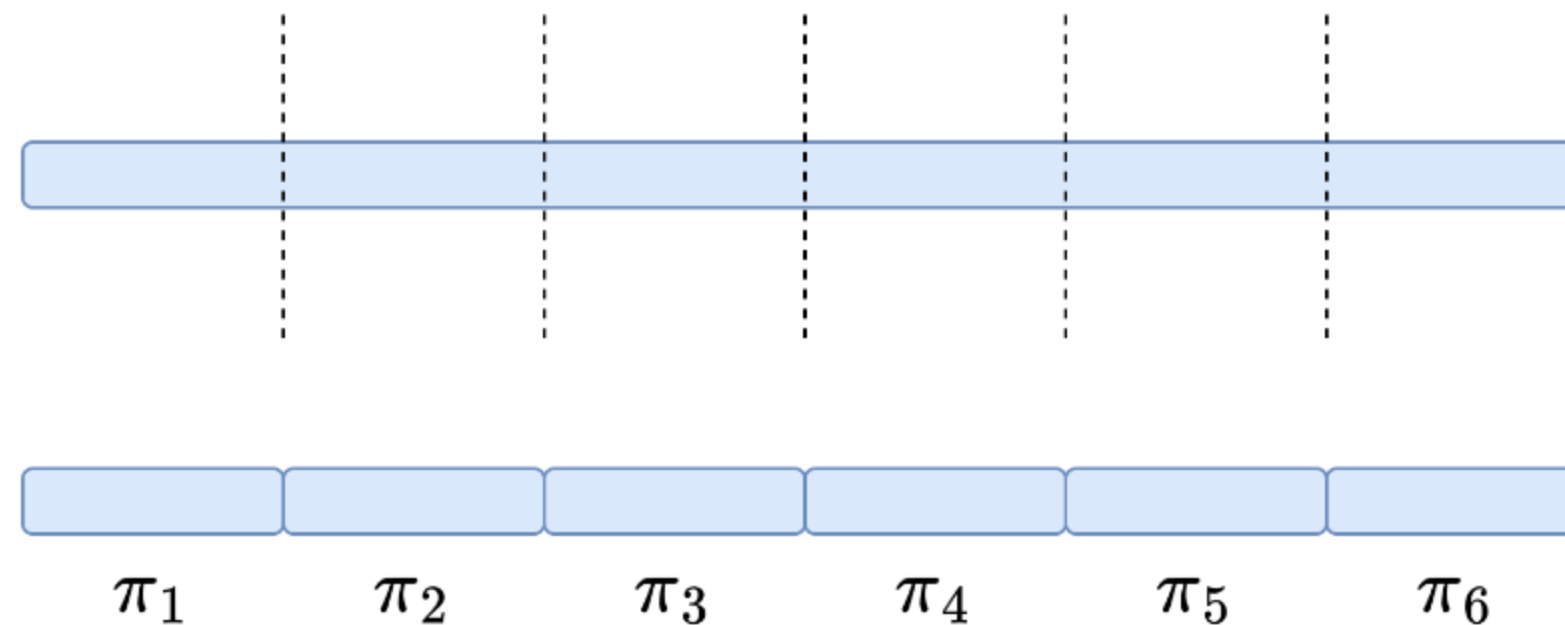
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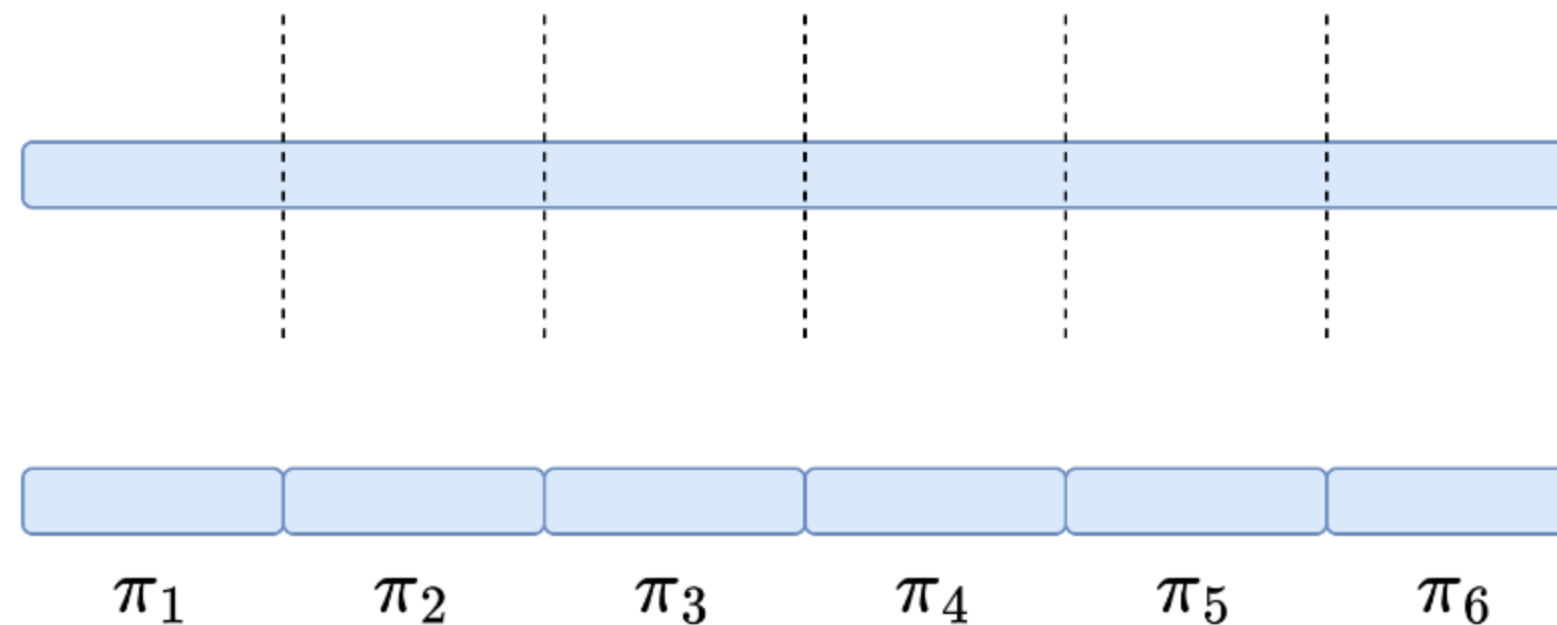
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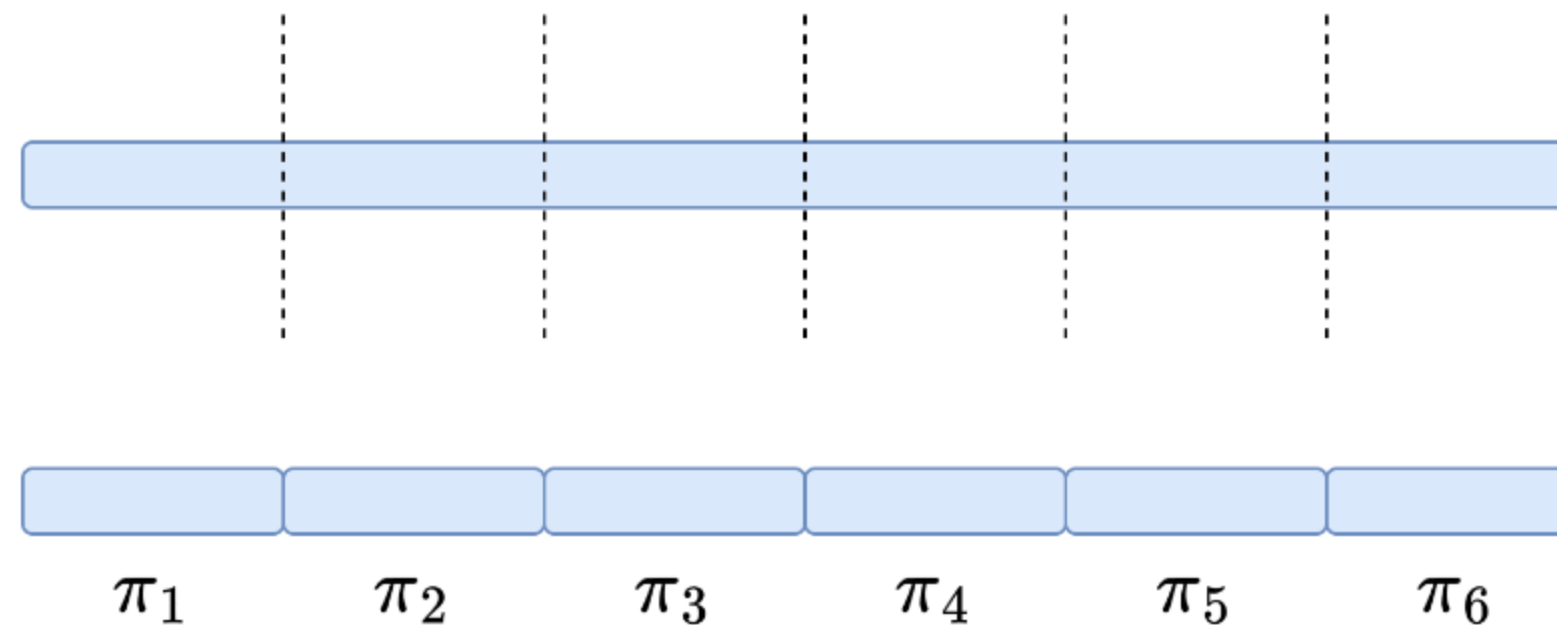


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Roadmap

~~1. Oblivious Shuffle Preprocessing~~

2. Scaling with Many Parties

Scaling with Many Parties

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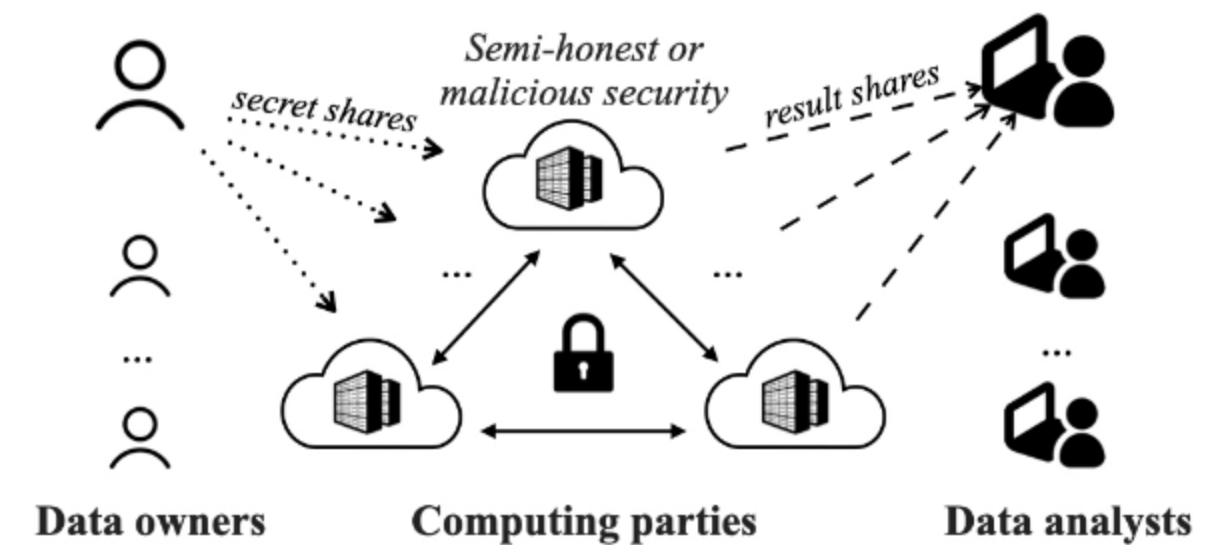
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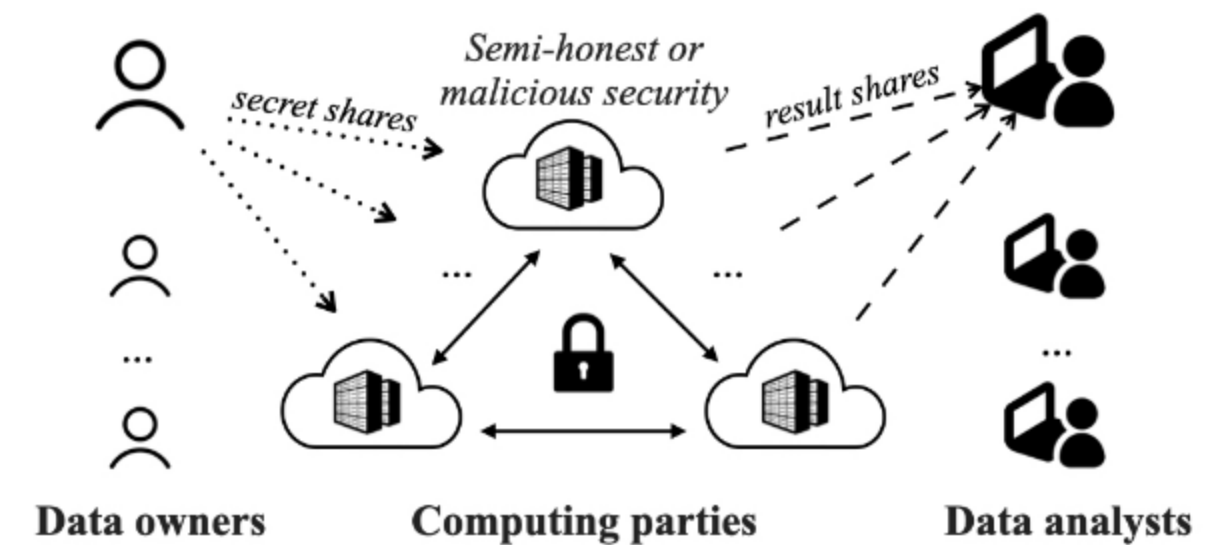
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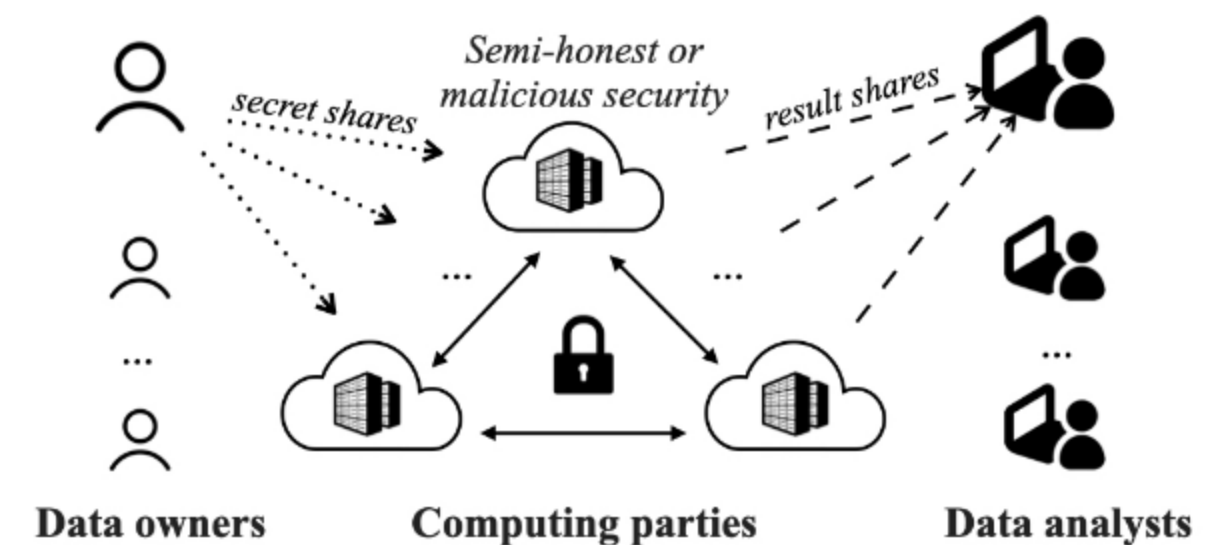
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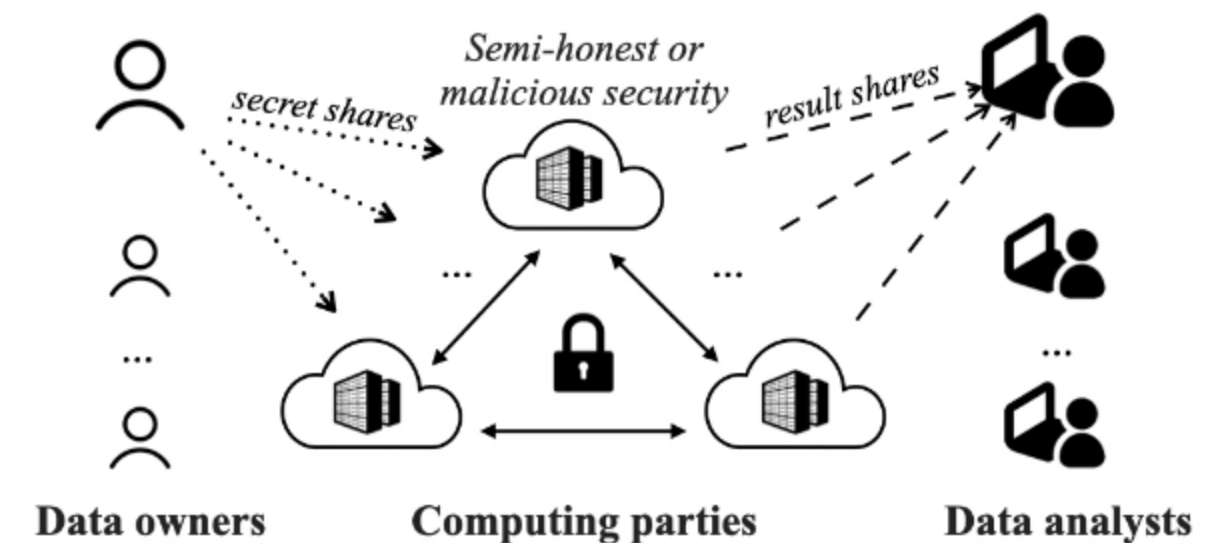
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Goal: Scale *practical* MPC to thousands of parties.

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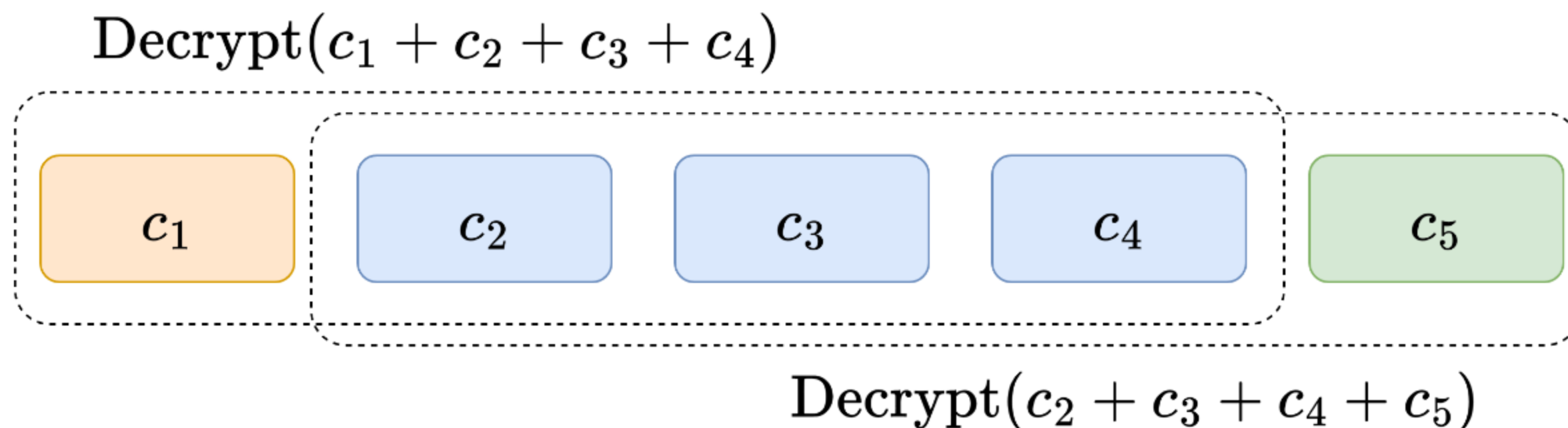
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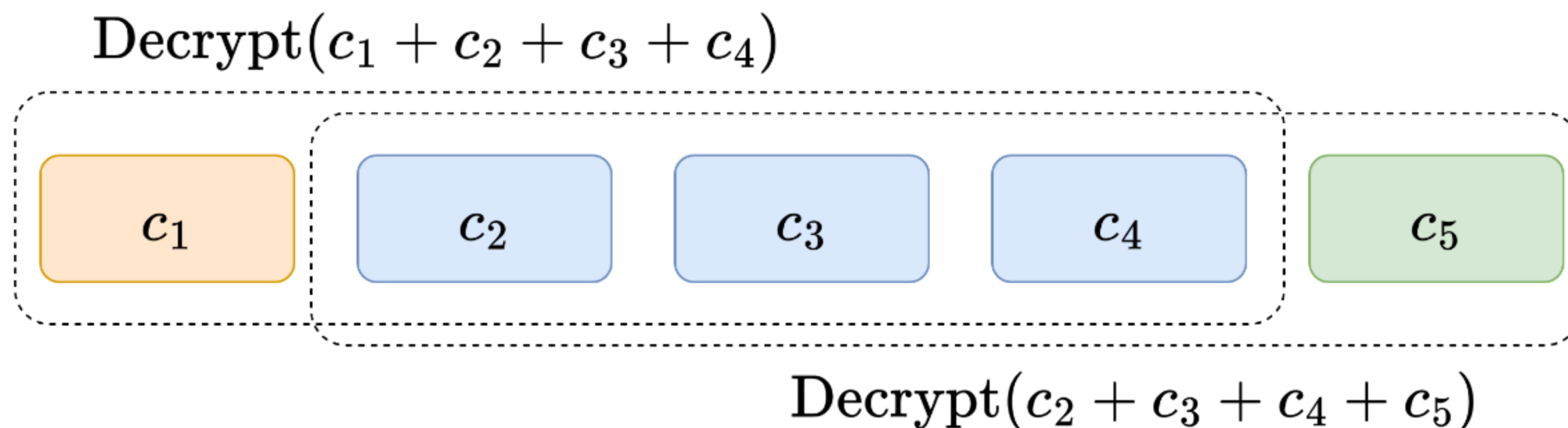


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The emerging paradigm uses a small committee to atomically decrypt (OPA, Willow, TACITA).



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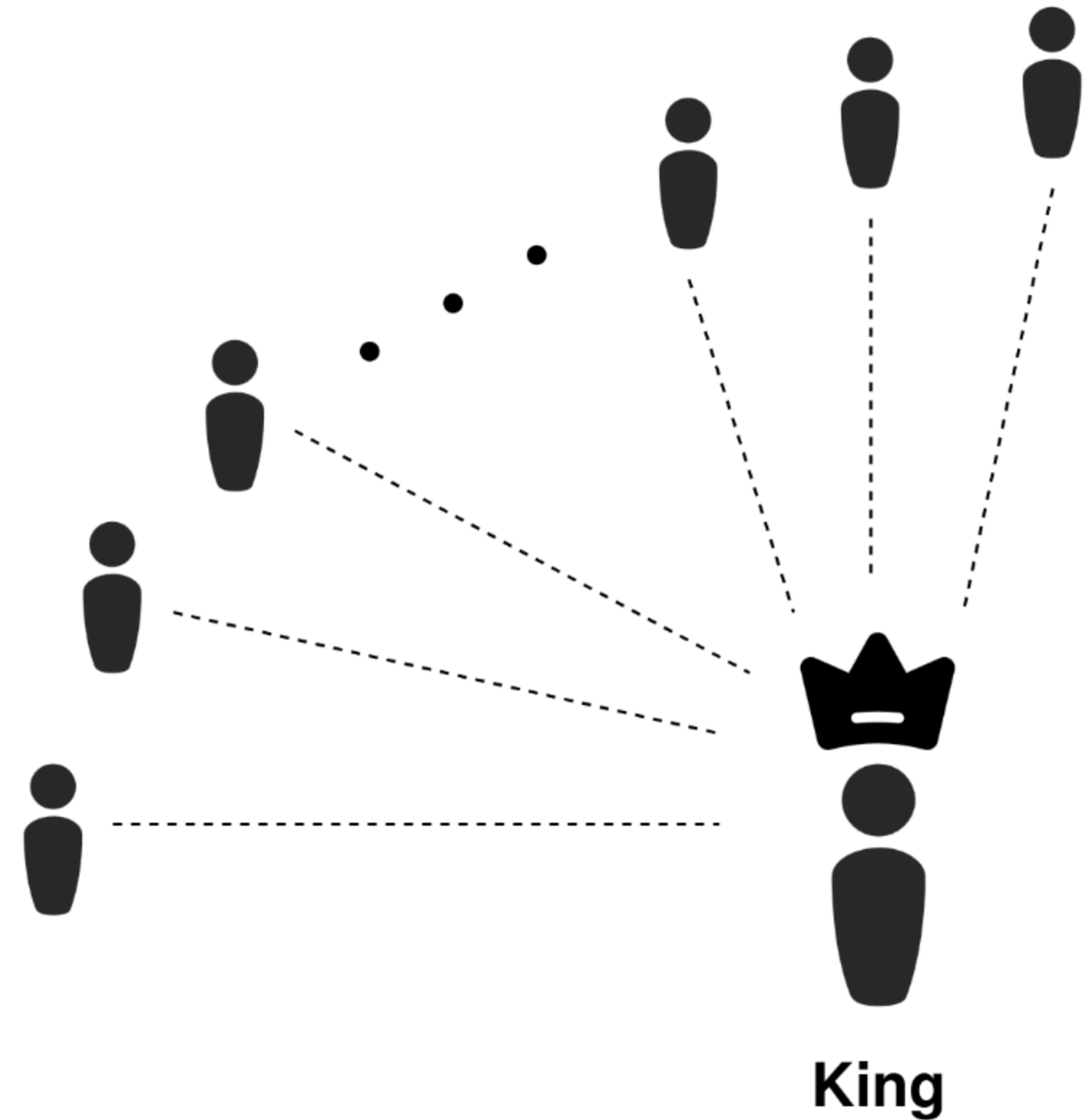
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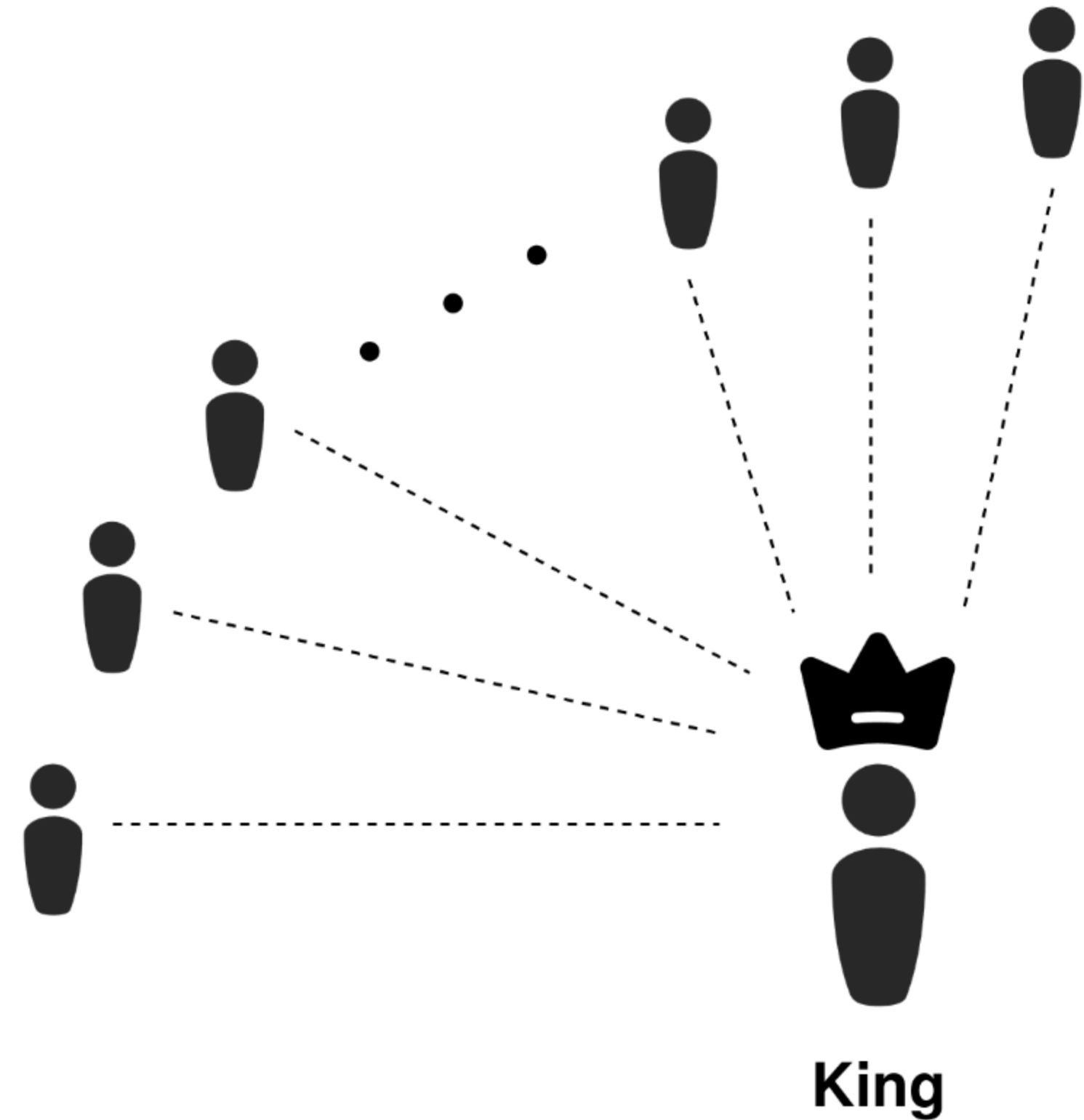
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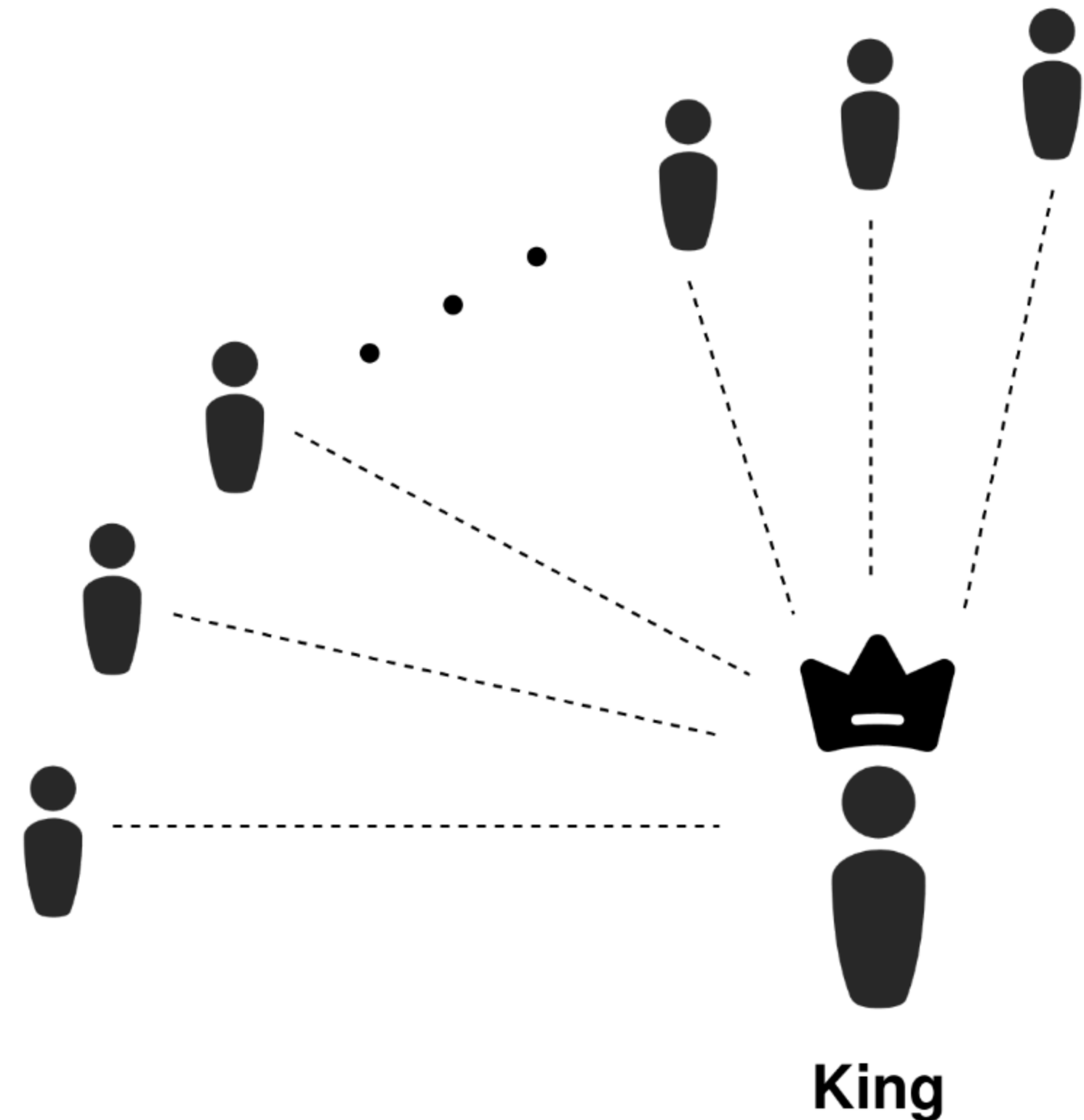
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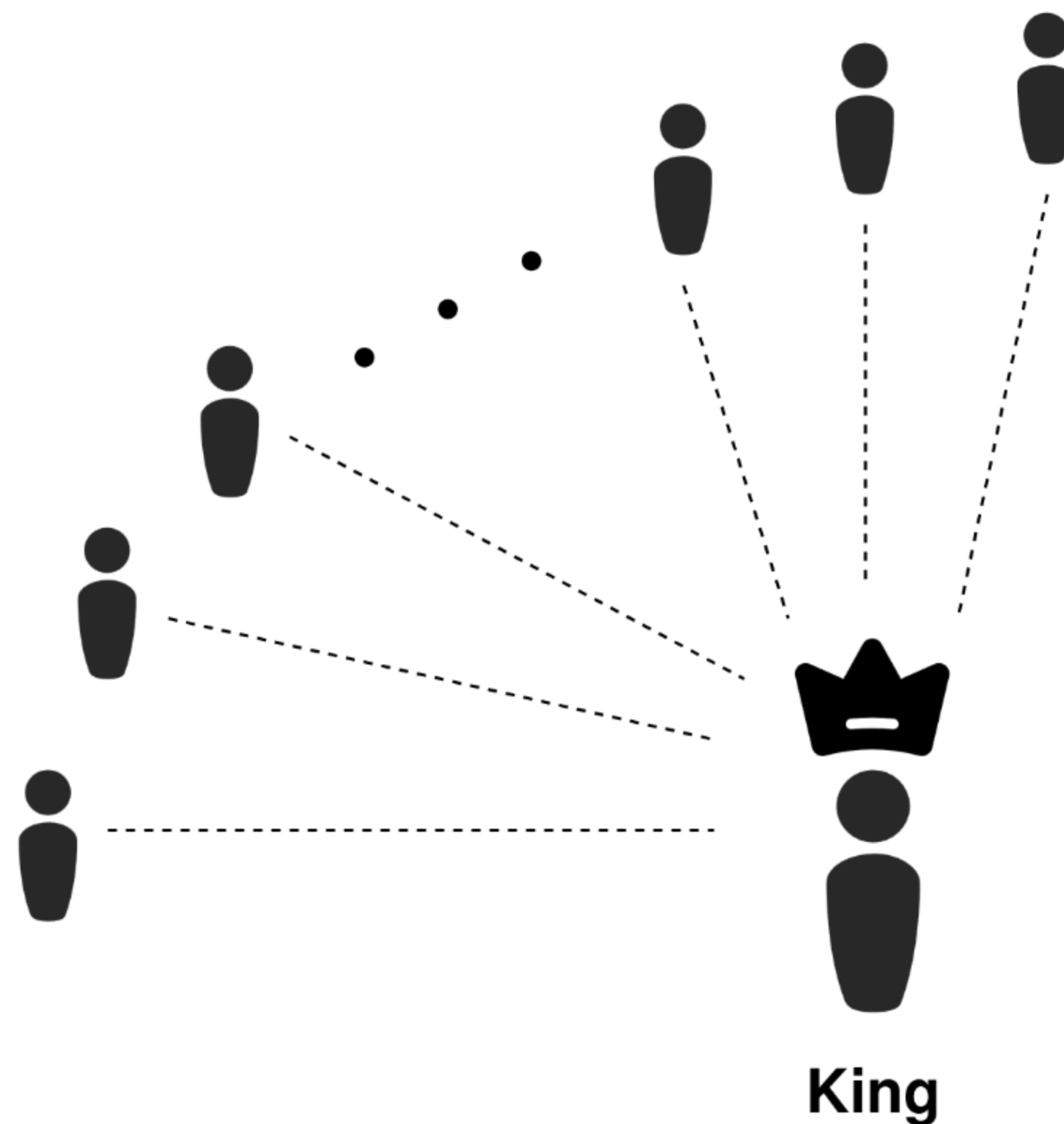
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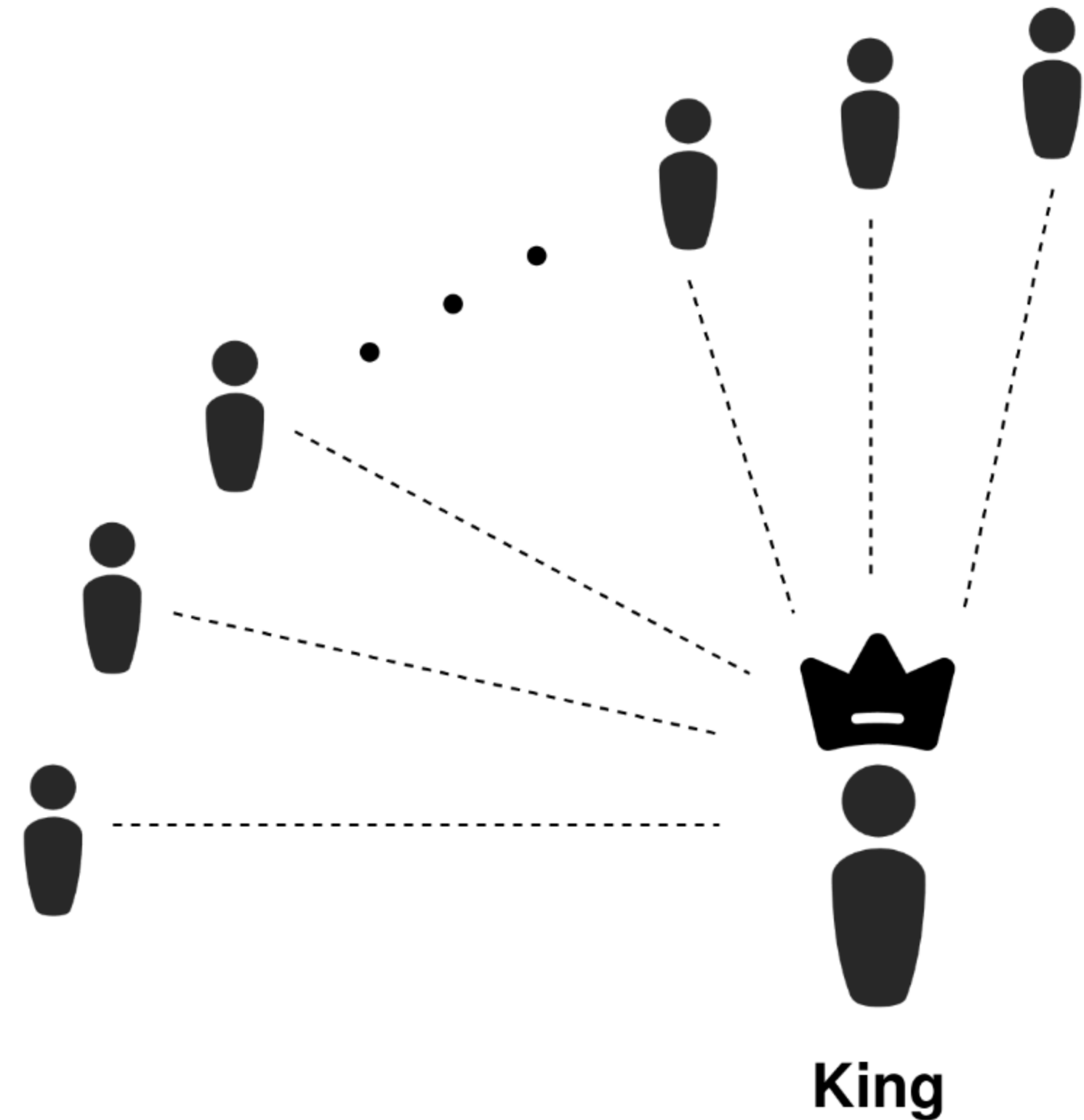
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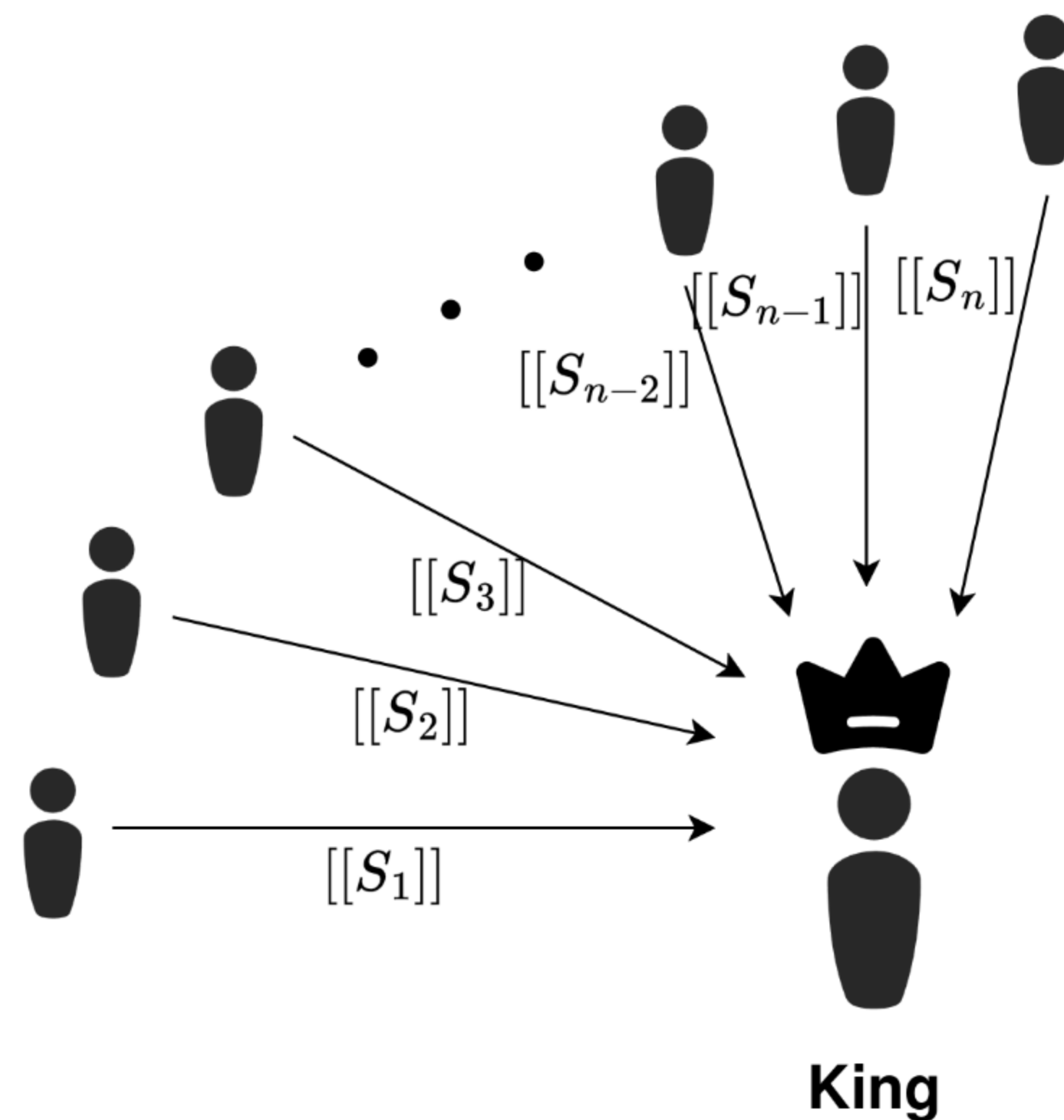
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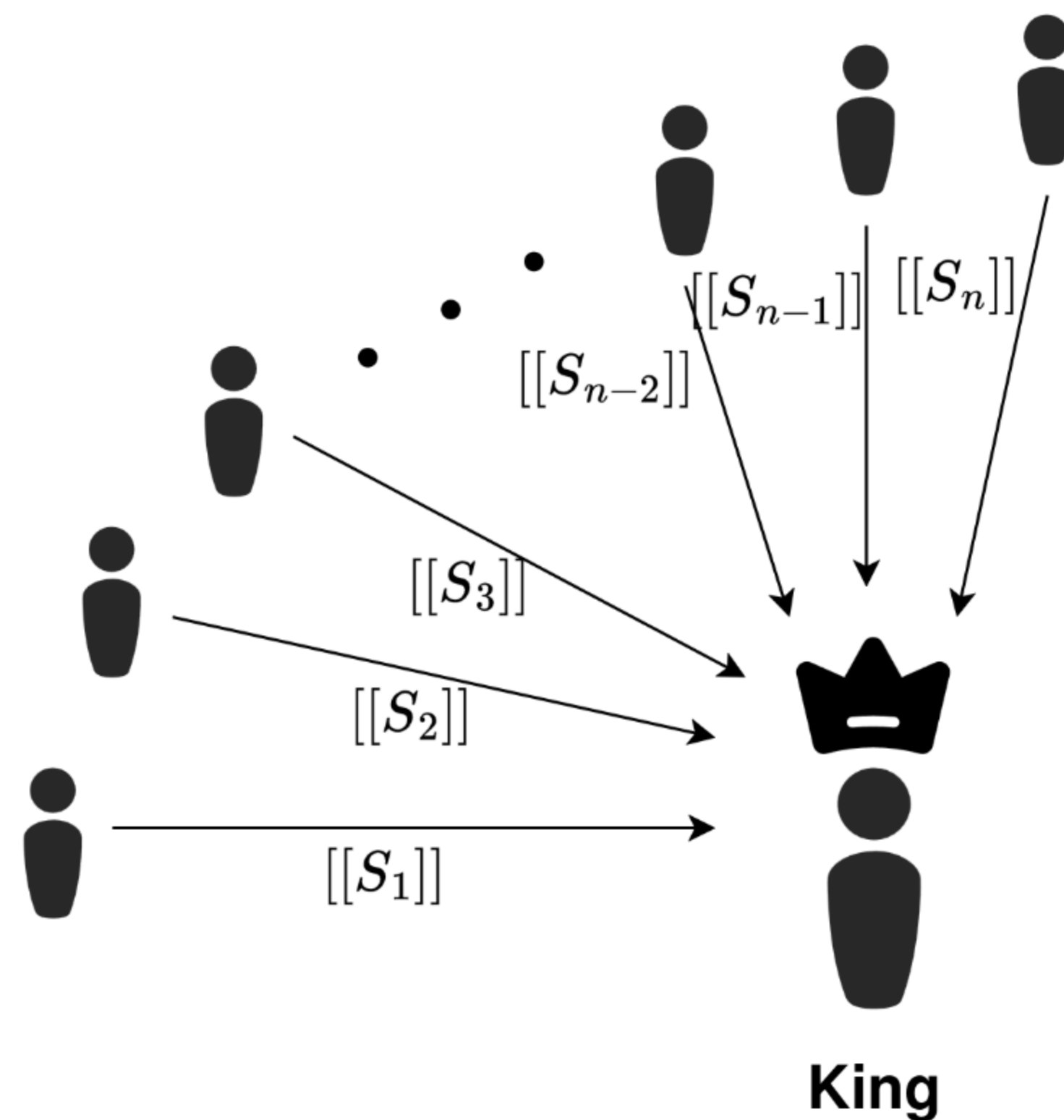
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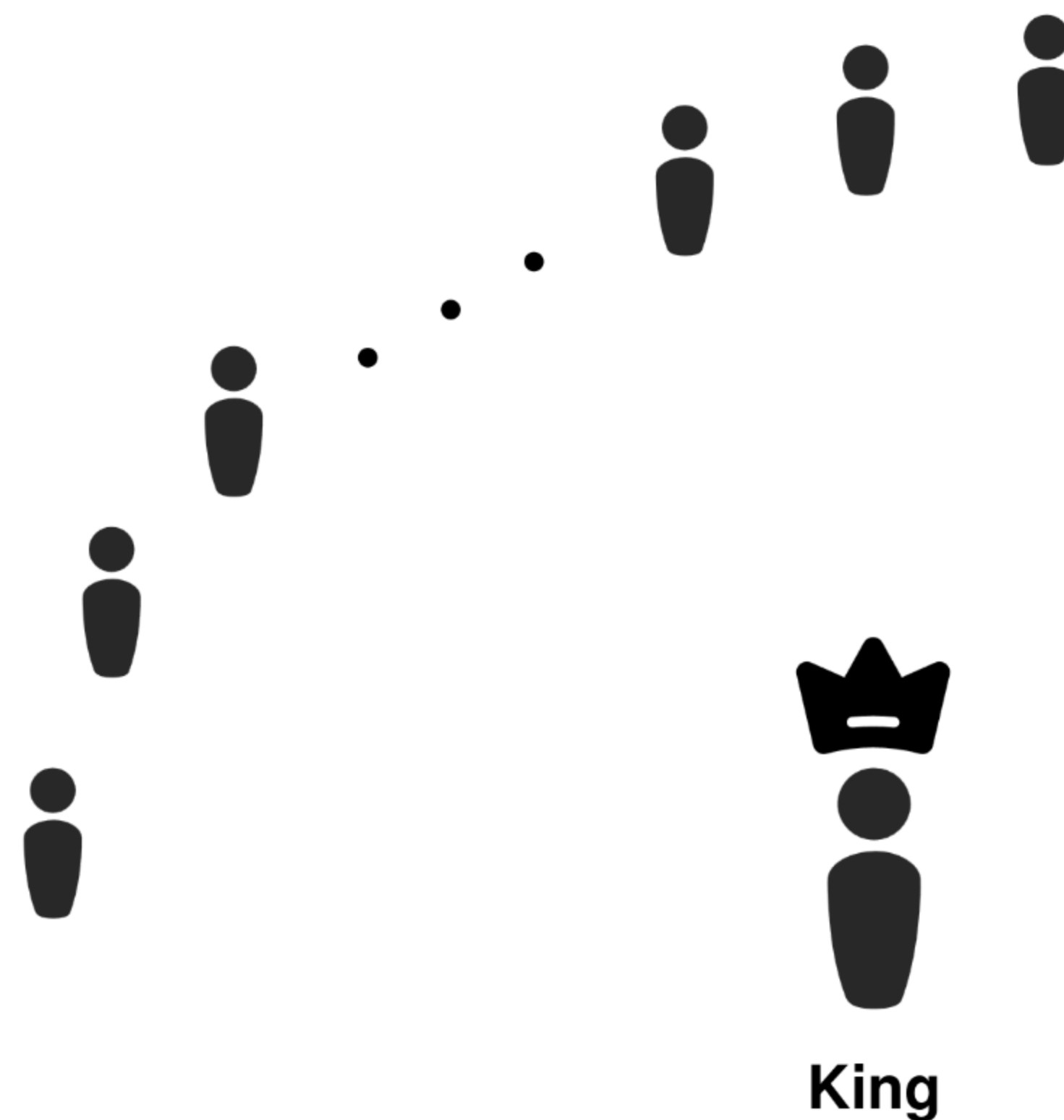
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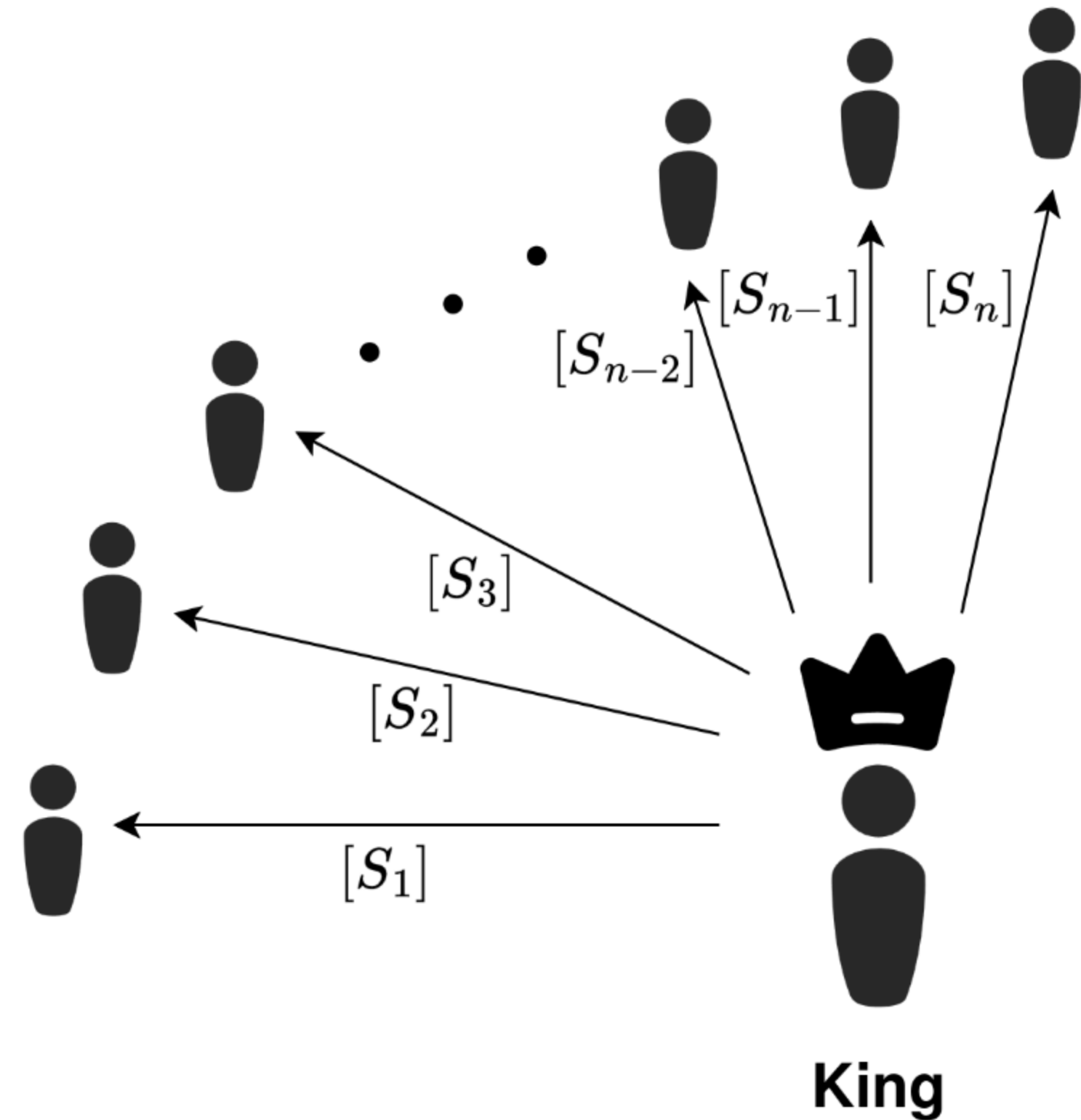
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- Targets machine learning inference
- Optimizes for round complexity
- Shows results with up to 63 parties

Scalable Multi-Party Computation Protocols for Machine Learning in the Honest-Majority Setting

Fengrun Liu
*University of Science and Technology of China &
Shanghai Qi Zhi Institute*

Xiang Xie
Shanghai Qi Zhi Institute & PADO Labs

Yu Yu
*Shanghai Jiao Tong University &
State Key Laboratory of Cryptology, P. O. Box 5159, Beijing, 100878, China*

Abstract

In this paper, we present a novel and scalable multi-party computation (MPC) protocol tailored for privacy-preserving machine learning (PPML) with semi-honest security in the honest-majority setting. Our protocol utilizes the Damgård-Nielsen (DN07) protocol with Mersenne prime fields.

Secure Multi-Party Computation (MPC) [3, 21, 47] enables a group of n parties to collectively compute a function on their private inputs while preserving the privacy of the inputs. MPC offers a promising approach to address these privacy challenges in ML. It enables privacy-preserving machine learning (PPML) by performing secure computations on dis-

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The paper simultaneously minimizes total work and makes base operations constant-round.

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- Latency constrained

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WAN Setting

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- Uses 5% of available bandwidth

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Proposal: Lean into the king as the bottleneck.

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- GPU acceleration



Future: Tolerating Dropouts

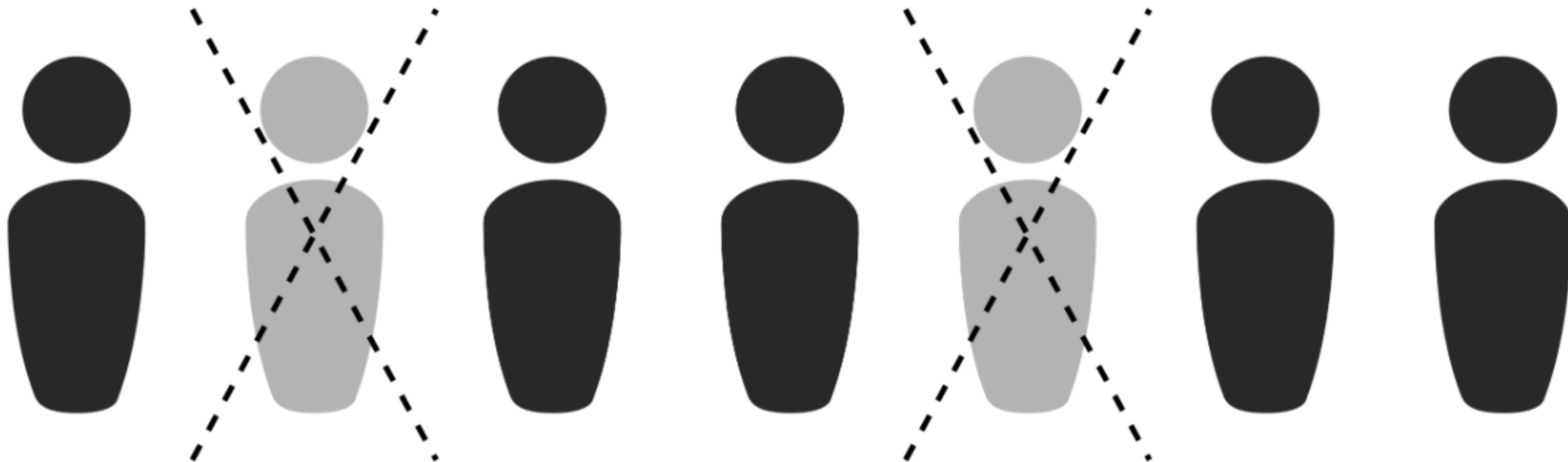
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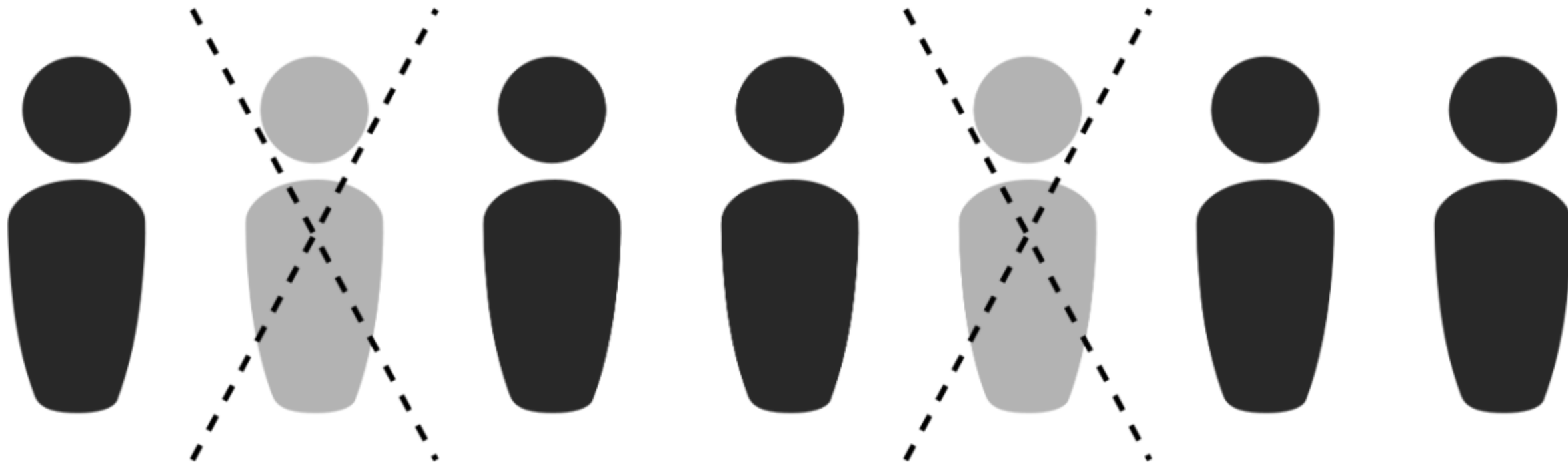


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- Subset of Fluid MPC

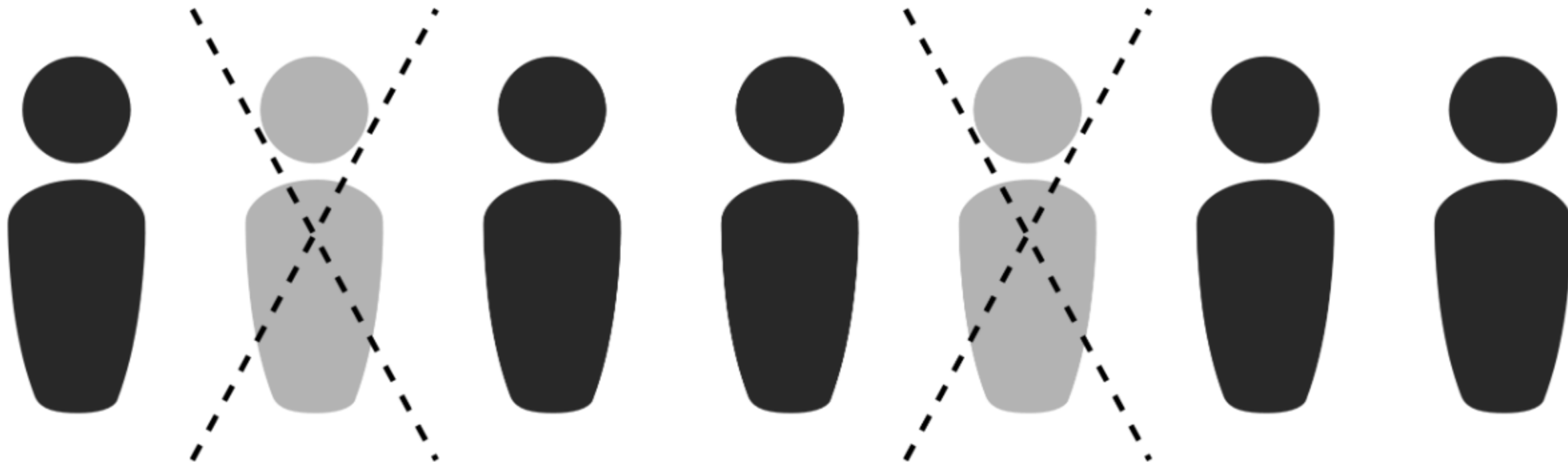


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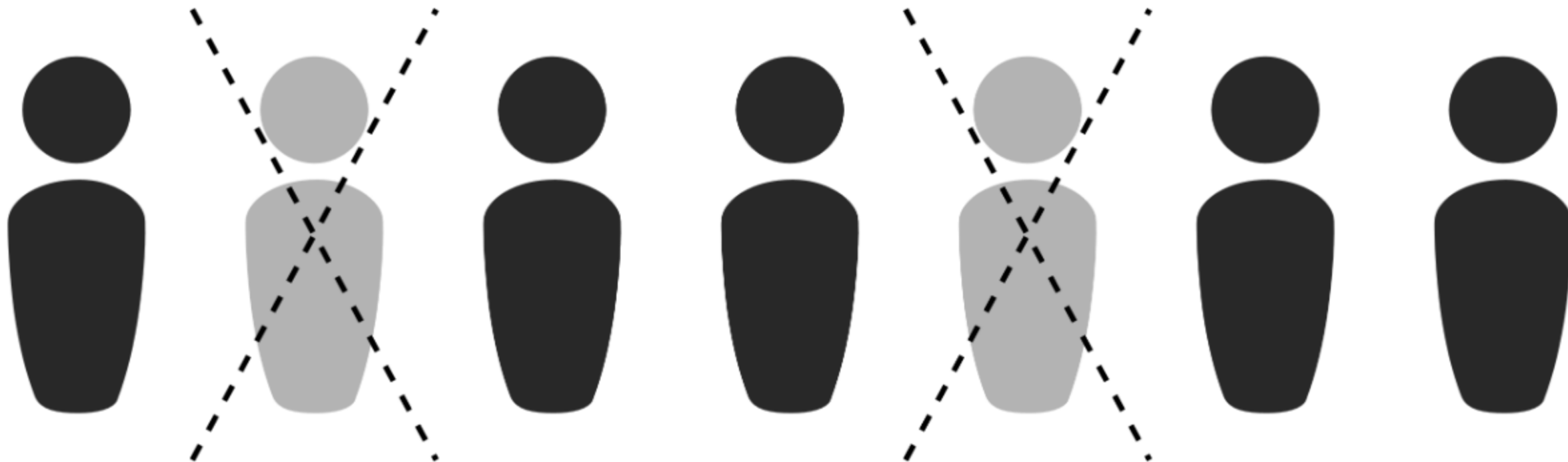


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- Semi-honest security is straightforward
- Malicious security is more complex



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Thank You!